

TRIGONOMETRIC EQUATIONS

WORKEDOUT EXAMPLES:

W.E-1: Solutions of the equation

$$\sin^4 x + \cos^4 x = \frac{7}{2} \sin x \cos x.$$

Ans: $x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}$ where $n \in \mathbb{Z}$

Sol:

$$\sin^4 x + \cos^4 x = \sin^4 x + 2 \sin^2 x \cos^2 x + \cos^4 x - 2 \sin^2 x \cos^2 x$$

$$= (\sin^2 x + \cos^2 x)^2 - 2 \sin^2 x \cos^2 x$$

where we get

$$\sin^4 x + \cos^4 x = 1 - \frac{1}{2} \sin^2 2x \quad \dots (i)$$

Using (i), the given equation becomes

$$1 - \frac{1}{2} \sin^2 2x = \frac{7}{4} \sin 2x$$

$$2 \sin^2 2x + 7 \sin 2x - 4 = 0$$

$$\Rightarrow (2 \sin 2x - 1)(\sin 2x + 4) = 0$$

either

$$2 \sin 2x - 1 = 0 \Rightarrow 2x = n\pi + (-1)^n \frac{\pi}{6} \text{ where } n \in \mathbb{Z}$$

or $\sin 2x + 4 = 0$ which has no solution because

$-4 \notin \text{range of } \sin 2x \text{ where } n \in \mathbb{Z}$

So, the solution of equation is

$$x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12} \text{ where } n \in \mathbb{Z}.$$

W.E-2: Solutions of the equation

$$3\sqrt{3} \sin^3 x + \cos^3 x + 3\sqrt{3} \sin x \cos x = 1.$$

Ans: $x = (2k+1)\pi - \frac{\pi}{3} \cup x = 2m\pi; m \in I, k \in I$

Sol: Let $a = \sqrt{3} \sin x$; $b = \cos x$; $c = -1$ then given equation reduces to

$$a^3 + b^3 + c^3 - 3abc = 0$$

$$\Rightarrow (a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 0$$

$$\Rightarrow (\sqrt{3} \sin x + \cos x - 1)$$

$$(3 \sin^2 x + \cos^2 x + 1 - \sqrt{3} \sin x \cos x - \cos x - \sqrt{3} \sin x) = 0.$$

Thus the given equation has solution if

$$\sqrt{3} \sin x + \cos x - 1 = 0$$

$$\Rightarrow \sqrt{3} \sin x + \cos x = 1 \Rightarrow \frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x = \frac{1}{2}$$

$$\Rightarrow \sin \left(x + \frac{\pi}{6} \right) = \sin \frac{\pi}{6}$$

$$\Rightarrow \left(x + \frac{\pi}{6} \right) = n\pi + (-1)^n \frac{\pi}{6}.$$

If n is odd;

$$\Rightarrow x = n\pi - \frac{\pi}{6} - \frac{\pi}{6} = n\pi - \frac{\pi}{3} \Rightarrow x = (2k+1)\pi - \frac{\pi}{3}$$

If n is even

$$\Rightarrow x = n\pi + \frac{\pi}{6} - \frac{\pi}{6} = n\pi \Rightarrow x = 2m\pi; m \in I.$$

W.E-3: Solution set of the equation

$$\log_{\frac{-x^2-6x}{10}} (\sin 3x + \sin x) = \log_{\frac{-x^2-6x}{10}} (\sin 2x).$$

Ans: $-\frac{5\pi}{3}$

Sol: $\log_{\frac{-x^2-6x}{10}} (\sin 3x + \sin x) = \log_{\frac{-x^2-6x}{10}} (\sin 2x)$

$$\sin 3x + \sin x > 0 \text{ and } \sin 2x > 0$$

$$2 \sin 2x \cos x > 0 \text{ and}$$

$$x \in I^{st} \text{ quadrant.}$$

$$\log_{\frac{-x^2-6x}{10}} \left(\frac{\sin 3x + \sin x}{\sin 2x} \right) = 0$$

$$\text{or } \frac{2 \sin 2x \cos x}{\sin 2x} = 1 \text{ or } 2 \cos x = 1$$

$$\text{or } \cos x = 1/2 \Rightarrow x = 2n\pi \pm \pi/3$$

$$\Rightarrow x \in (-6, 0) \left(\because \frac{-x^2 - 6x}{10} > 0 \text{ and } \frac{-x^2 - 6x}{10} \neq 1 \right).$$

$$\therefore \text{for } n=-1 \Rightarrow x = -2\pi + \frac{\pi}{3} = \frac{-5\pi}{3}$$

W.E-4: Solution of the equation

$$\sin\left(\frac{\sqrt{x}}{2}\right) + \cos\left(\frac{\sqrt{x}}{2}\right) = \sqrt{2} \sin \sqrt{x}$$

$$\text{Ans: } x = \left(n\pi + (-1)^{n+1} \frac{\pi}{6}\right)^2 \text{ or } x = \left(n\pi + (-1)^n \frac{\pi}{2}\right)^2$$

$$\text{Sol: } \sin\left(\frac{\sqrt{x}}{2}\right) + \cos\left(\frac{\sqrt{x}}{2}\right) = \sqrt{2} \sin \sqrt{x}$$

$$1 + 2 \sin \frac{\sqrt{x}}{2} \cos \frac{\sqrt{x}}{2} = 2 \sin^2 \sqrt{x}$$

$$1 + \sin \sqrt{x} = 2 \sin^2 \sqrt{x}$$

$$2 \sin^2 \sqrt{x} - \sin \sqrt{x} - 1 = 0$$

$$2 \sin^2 \sqrt{x} - 2 \sin \sqrt{x} + \sin \sqrt{x} - 1 = 0$$

$$2 \sin \sqrt{x} (\sin \sqrt{x} - 1) + 1 (\sin \sqrt{x} - 1) = 0$$

$$\sin \sqrt{x} = -\frac{1}{2} \text{ or } \sin \sqrt{x} = 1$$

$$\sqrt{x} = n\pi + (-1)^n \left(\frac{-\pi}{6}\right); \sqrt{x} = n\pi + (-1)^n \frac{\pi}{2}$$

$$x = \left(n\pi + (-1)^{n+1} \frac{\pi}{6}\right)^2 \text{ or } x = \left(n\pi + (-1)^n \frac{\pi}{2}\right)^2$$

W.E-5: The least positive angle measured in degrees satisfying the equation

$$\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$$

$$\text{Ans: } 72^\circ$$

Sol:

$$\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$$

$$\left(\because (a+b+c)^3 = a^3 + b^3 + c^3\right) \Rightarrow (a+b)(b+c)(c+a) = 0$$

$$\Rightarrow (\sin x + \sin 2x)(\sin 2x + \sin 3x)(\sin 3x + \sin x) = 0$$

$$\Rightarrow 2 \sin \frac{3x}{2} \cos \frac{x}{2} = 0 \text{ (or) } 2 \sin \frac{5x}{2} \cos \frac{x}{2} = 0 \text{ (or)}$$

$$2 \sin 2x \cos x = 0$$

$$\Rightarrow \sin \frac{3x}{2} = 0 \Rightarrow x = \frac{2n\pi}{3}; n \in \mathbb{Z}$$

$$\text{or } \sin \frac{5x}{2} = 0 \Rightarrow x = \frac{2m\pi}{5}; m \in \mathbb{Z}$$

$$\text{or } \sin 2x = 0 \Rightarrow x = \frac{n\pi}{2}; k \in \mathbb{Z}$$

$$\text{or } \cos \frac{x}{2} = 0 \Rightarrow x = (2p+1)\pi; p \in \mathbb{Z}$$

$$\text{or } \cos x = 0 \Rightarrow x = (2q+1)\frac{\pi}{2}; q \in \mathbb{Z}.$$

$$\therefore \text{Least positive angle is } \frac{2\pi}{5} = 72^\circ.$$

W.E-6: The set of value of 'a' for which the equation

$$\sin^4 x + \cos^4 x + \sin 2x + a = 0 \text{ possesses solutions.}$$

$$\text{Ans: } a \in \left[-\frac{3}{2}, \frac{1}{2}\right]$$

$$\text{Sol: } \sin^4 x + \cos^4 x + \sin 2x + a = 0.$$

$$(\sin^2 x)^2 + (\cos^2 x)^2 + 2 \sin^2 x \cos^2 x - 2 \sin^2 x \cos^2 x + 2 \sin x \cos x + a = 0$$

$$1 - \frac{\sin^2 2x}{2} + \sin 2x + a = 0$$

$$2 - \sin^2 2x + 2 \sin 2x + 2a = 0$$

$$\text{or } 2a = \sin^2 2x - 2 \sin 2x - 2$$

$$\text{or } 2a + 3 = (\sin 2x - 1)^2$$

$$\Rightarrow 2a + 3 = (y-1)^2 \text{ where } y \in [-1, 1]$$

$$\Rightarrow 2a + 3 \in [0, 4]$$

$$\Rightarrow a \in \left[-\frac{3}{2}, \frac{1}{2}\right]$$

$$\sin 2x = 1 - \sqrt{2a+3}$$

$$\sin 2x = \sin \alpha \text{ where}$$

$$\alpha = \sin^{-1} \left(1 - \sqrt{2a+3} \right)$$

$$\text{or } x = \frac{1}{2} \left[n\pi + (-1)^n \sin^{-1} \left(1 - \sqrt{2a+3} \right) \right]$$

$$\text{where } a \in \left[-\frac{3}{2}, \frac{1}{2} \right].$$

W.E-7: The general solution of the equation,

$$\sin \frac{2x+1}{x} + \sin \frac{2x+1}{3x} - 3 \cos^2 \frac{2x+1}{3x} = 0.$$

$$\text{Ans: } x = \frac{1}{3\alpha-2} \text{ where } \alpha = n\pi + (-1)^n \sin^{-1} \left(\frac{3}{4} \right)$$

$$\text{Sol: } \sin \frac{2x+1}{x} + \sin \frac{2x+1}{3x} - 3 \cos^2 \frac{2x+1}{3x} = 0$$

$$2 \sin \frac{4(2x+1)}{2 \times 3x} \cos \frac{2x+1}{3x} = 3 \cos^2 \frac{2x+1}{3x}$$

$$\text{or } 2 \sin \frac{2(2x+1)}{3x} \cos \frac{2x+1}{3x} = 3 \cos^2 \frac{2x+1}{3x}$$

or

$$4 \sin \left(\frac{2x+1}{3x} \right) \cos^2 \left(\frac{2x+1}{3x} \right) - 3 \cos^2 \frac{2x+1}{3x} = 0$$

$$\Rightarrow \cos^2 \left(\frac{2x+1}{3x} \right) = 0 \text{ (or)}$$

$$\sin \frac{2x+1}{3x} = \frac{3}{4} = \sin \alpha, \text{ where } \alpha = n\pi + (-1)^n \sin^{-1} \left(\frac{3}{4} \right)$$

$$\cos^2 \left(\frac{3x+1}{3x} \right) = \cos^2 \frac{\pi}{2}$$

$$\Rightarrow \frac{2x+1}{3x} = n\pi \pm \frac{\pi}{2}$$

$$\Rightarrow x = \frac{1}{3(2n \pm 1)\pi - 2}$$

$$\text{or } \sin \left(\frac{2x+1}{3x} \right) = \sin \alpha$$

$$\Rightarrow x = \frac{1}{3\alpha-2} \text{ where } \alpha = n\pi + (-1)^n \sin^{-1} \left(\frac{3}{4} \right)$$

W.E-8: Solve the equation

$$\begin{aligned} & \cos(3^x \pi) - 2 \cos^2(3^x \pi) + 2 \cos(4(3^x) \pi) - \cos(7(3^x) \pi) \\ &= \sin(3^x \pi) + 2 \sin^2(3^x \pi) - 2 \sin(4(3^x) \pi) \\ & \quad + 2 \sin(3^{x+1} \pi) - \sin(7(3^x) \pi) \end{aligned}$$

$$\text{Ans: } x = \log_3 \left(\frac{n}{4} + (-1)^n \frac{\pi}{16} - \frac{\pi}{16} \right), n \in \mathbb{Z}$$

Sol: Let $3^x \pi = \theta$ then

$$\begin{aligned} & \Rightarrow \cos \theta - 2 \cos^2 \theta + 2 \cos 4\theta - \cos 7\theta \\ &= \sin \theta + 2 \sin^2 \theta - 2 \sin 4\theta + 2 \sin 3\theta - \sin 7\theta \end{aligned}$$

$$\begin{aligned} & \Rightarrow \cos \theta - \cos 7\theta + \sin 7\theta - \sin \theta \\ & \quad + 2(\cos 4\theta + \sin 4\theta) = 2 + 2 \sin 3\theta \end{aligned}$$

$$\begin{aligned} & \Rightarrow 2 \sin 4\theta \sin 3\theta + 2 \cos 4\theta \sin 3\theta \\ & \quad + 2(\cos 4\theta + \sin 4\theta) = 2 + 2 \sin 3\theta \end{aligned}$$

$$\begin{aligned} & \Rightarrow \sin 4\theta (\sin 3\theta + 1) + \cos 4\theta (\sin 3\theta + 1) \\ & \quad - (1 + \sin 3\theta) = 0 \end{aligned}$$

$$\Rightarrow (\sin 3\theta + 1)(\sin 4\theta + \cos 4\theta - 1) = 0$$

$$\Rightarrow \sin 3\theta = -1 \Rightarrow 3\theta = 2n\pi - \frac{\pi}{2}$$

$$\Rightarrow 3\pi 3^x = 2n\pi - \pi/2$$

$$\Rightarrow 3^{x+1} = 2n - \frac{1}{2} \Rightarrow x+1 = \log_3 \frac{4n-1}{2}$$

$$\Rightarrow x = \log_3 \frac{4n-1}{2} - 1$$

$$\text{And } \sin 4\theta + \cos 4\theta = 1$$

$$\Rightarrow \sin \left(4\theta + \frac{\pi}{4} \right) = \sin \frac{\pi}{4} \Rightarrow 4\theta + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}$$

$$\Rightarrow 4\pi 3^x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$$

$$\Rightarrow 3^x = \frac{n}{4} + (-1)^n \frac{\pi}{16} - \frac{\pi}{16}$$

$$\Rightarrow x = \log_3 \left(\frac{n}{4} + (-1)^n \frac{\pi}{16} - \frac{\pi}{16} \right), n \in \mathbb{Z}$$

W.E-9: The value of 'a' so that the equation

$$\cos^2 \left[\frac{\pi}{10} (2 \cos^2 x - 5 \cos x + 2) \right] = \sec^2 (x + a \sec^2 x)$$

has a real solution.

Ans: $a = \frac{r\pi}{2} \pm \frac{\pi}{6}$ where $r \in \mathbb{I}$

Sol: Since $\cos^2 \theta \leq 1$ and $\sec^2 \alpha \geq 1$

$$\Rightarrow \cos^2 \left[\frac{\pi}{10} (2 \cos^2 x - 5 \cos x + 2) \right] = 1$$

$$\Rightarrow \frac{\pi}{10} (2 \cos^2 x - 5 \cos x + 2) = 2n\pi \quad \forall n \in \mathbb{I}$$

$$\Rightarrow 2 \cos^2 x - 5 \cos x + 2 = 20n, n \in \mathbb{I}.$$

Now

$$2 \cos^2 x - 5 \cos x + 2 = 2 \left[\cos x - \frac{5}{4} \right]^2 - \frac{9}{8}$$

$$\Rightarrow -1 \leq 2 \cos^2 x - 5 \cos x + 2 \leq 9$$

$\Rightarrow$ In equation (1) only one value of n is possible i.e., $n = 0$

$$\Rightarrow 2 \cos^2 x - 5 \cos x + 2 = 0$$

$$\Rightarrow (2 \cos x - 1)(\cos x - 2) = 0$$

$$\Rightarrow x = 2n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow \sec^2 \left(2n\pi \pm \frac{\pi}{3} + a \sec^2 \left(2n\pi \pm \frac{\pi}{3} \right) \right) = 1$$

$$\Rightarrow \sec^2 \left(2n\pi \pm \frac{\pi}{3} + 2a \right) = 1 \Rightarrow 2a = r\pi \pm \frac{\pi}{3}$$

$$\Rightarrow a = \frac{r\pi}{2} \pm \frac{\pi}{6} \text{ where } r \in \mathbb{I}.$$

W.E-10: The number of roots of the equation,

$$\sin x + 2 \sin 2x = 3 + \sin 3x \text{ is } \quad \text{[IIT 2014]}$$

- (A) 0 (B) 1
(C) 2 (D) infinite

Ans: A

Sol: $(\sin x - \sin 3x) + 2 \sin 2x = 3$

$$\Rightarrow -2 \sin x \cos 2x + 2 \sin 2x = 3$$

$$\Rightarrow 2 \sin x \cos 2x - 2 \sin 2x + 3 = 0$$

$$\Rightarrow 2 \sin x \cos 2x - 2 \sin 2x + (\sin^2 x + \cos^2 x) + (\sin^2 2x + \cos^2 2x) + 1 = 0$$

$$\Rightarrow (1 - \sin 2x)^2 + (\sin x + \cos 2x)^2 + \cos^2 x = 0$$

Hence there is no root because all terms do not vanish simultaneously.

W.E-11: Total number of solutions of

$$[\sin x] + \cos x = 0 \text{ for } x \in [0, 2\pi] \text{ is}$$

(where $[.]$ denotes the greatest integer function)

- (A) 1 (B) 2 (C) 4 (D) 0

Ans: D

Sol: Given equation is $[\sin x] = -\cos x$

Real solution exist iff $-\cos x$ is integer i.e., $\cos x$ is an integer.

So possible solutions for $x \in [0, 2\pi]$ are

$$x = 0, \pi/2, \pi, 3\pi/2, 2\pi.$$

when $x = 0, 2\pi$ RHS = -1 and LHS = 0 no solution

$x = \pi/2$ or $3\pi/2$ RHS = 0 and LHS = 1 no solution

$x = \pi$ RHS = 1 and LHS = 0 no solution.

So, we can conclude that equations has no real solution exist.

W.E-12: Number of solutions to the equation

$$|2e^{\sin x} - 3 - e^{2\sin x}| = |e^{\sin x} - e^{2\sin x} - 1| \text{ in } [0, 2\pi] \text{ is}$$

- (A) zero (B) two
(C) four (D) one

Ans: (B)

Sol: Let ' $e^{\sin x} = t$ ' then $|2t - 3 - t^2| = |t - t^2 - 1|$

$$\Rightarrow |t^2 - 2t + 3| = |t^2 - t + 1|$$

here

$$t^2 - 2t + 3 > 0, t^2 - t + 1 > 0 \quad t \in \mathbb{R},$$

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So,

$$t^2 - 2t + 3 = t^2 - t + 1 \Rightarrow t = 2 \Rightarrow e^{\sin x} = 2$$

$$\Rightarrow \sin x = \ln 2 > 0$$

so x has two solutions in $[0, 2\pi]$

W.E-13: Complete set of values of x in $(0, \pi)$

satisfying $1 + \log_2 \sin x + \log_2 \sin 3x \geq 0$ is

(A) $\left(\frac{2\pi}{3}, \frac{3\pi}{4}\right)$ (B) $\left(\frac{\pi}{3}, \frac{2\pi}{3}\right)$

(C) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \pi\right)$ (D) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

Ans: A

Sol: $1 + \log_2 \sin x + \log_2 \sin 3x \geq 0$

$$\log_2 (2 \sin x \cdot \sin 3x) \geq 0$$

$$\Rightarrow 2 \sin x \cdot \sin 3x \geq 1$$

For $\sin x > 0 \Rightarrow x \in (0, \pi)$; $\sin 3x > 0$

$$\Rightarrow 3x \in (0, \pi) \cup (2\pi, 3\pi)$$

$$\Rightarrow x \in \left(0, \frac{\pi}{3}\right) \cup \left(\frac{2\pi}{3}, \pi\right)$$

For $2 \sin x \sin 3x \geq 1$

$$\Rightarrow 2 \sin^2 x (3 - 4 \sin^2 x) \geq 1$$

$$\Rightarrow 8 \sin^4 x - 6 \sin^2 x + 1 \leq 0$$

$$\Rightarrow (2 \sin^2 x - 1)(4 \sin^2 x - 1) \leq 0$$

$$\Rightarrow \left(\sin x - \frac{1}{\sqrt{2}}\right)\left(\sin x + \frac{1}{\sqrt{2}}\right)$$

$$\left(\sin x - \frac{1}{2}\right)\left(\sin x + \frac{1}{2}\right) \leq 0$$

$$\Rightarrow \frac{1}{2} \leq \sin x \leq \frac{1}{\sqrt{2}}$$

$$\Rightarrow x \in \left[\frac{\pi}{3}, \frac{\pi}{4}\right] \cup \left[\frac{2\pi}{3}, \frac{2\pi}{4}\right]$$

Thus finally $x \in \left(\frac{2\pi}{3}, \frac{3\pi}{4}\right)$

W.E-14: If $\operatorname{cosec}(\theta - \alpha), \operatorname{cosec} \theta, \operatorname{cosec}(\theta + \alpha)$

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are in A.P and $\alpha \in (-\pi/2, \pi/2)$ then the number of possible values of α is

- (A) 0 (B) 1
(C) 2 (D) 4

Ans: B

Sol:

$$\frac{2}{\sin \theta} = \frac{1}{\sin(\theta - \alpha)} + \frac{1}{\sin(\theta + \alpha)} = \frac{2 \sin \theta \cos \alpha}{\sin^2 \theta - \sin^2 \alpha}$$

$$\Rightarrow \sin^2 \theta \cos \alpha = \sin^2 \theta - \sin^2 \alpha$$

$$\Rightarrow \sin^2 \theta = \frac{\sin^2 \alpha}{1 - \cos \alpha} = 2 \cos^2 \alpha / 2 \text{ if } \alpha \neq 0$$

$$\Rightarrow \cos^2 \alpha / 2 = 1/2 \sin^2 \theta \leq 1/2 - \pi/4 < \alpha/2 < \pi/4$$

$$\Rightarrow \cos^2(\alpha/2) \geq 1/2$$

$$\therefore \cos^2(\alpha/2) = 1/2 \Rightarrow \cos \alpha = 0$$

Which is not true in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

If $\alpha = 0$ then $\operatorname{cosec}(\theta - \alpha),$

$\operatorname{cosec} \theta, \operatorname{cosec}(\theta + \alpha)$ are obviously in A.P.

Hence $\alpha = 0$ is only possible value of α in

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

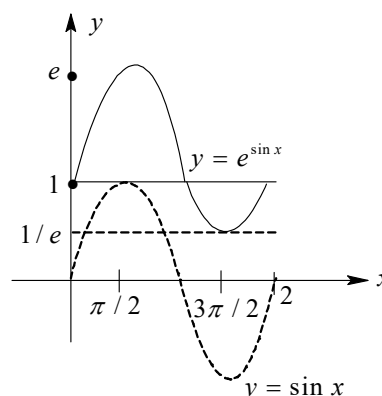
W.E-15: Find the number of solutions of the equation

$$4e^{\sin x} - 3e^{-\sin x} + 4 = 0 \text{ in } [0, 2\pi]$$

- (A) 1 (B) 2
(C) 3 (D) none of these

Ans: B

Sol: Let us suppose $e^{\sin x} = y$



$$\Rightarrow 4y - 3/y + 4 = 0$$

$$\Rightarrow 4y^2 + 4y - 3 = 0$$

$$\Rightarrow 4y^2 + 6y - 2y - 3 = 0$$

$$\Rightarrow (2y-1)(2y+3) = 0$$

$$\Rightarrow y = 1/2 \Rightarrow y = -3/2$$

$$\Rightarrow e^{\sin x} = 1/2$$

$$\Rightarrow -1 \leq \sin x \leq 1; 1/e \leq e^{\sin x} \leq e.$$

$e^{\sin x}$ can never be negative hence no solution

for $y = -3/2$ and $1/2 > 1/e$

$$\Rightarrow \text{No. of solutions} = 2.$$

W.E-16: If

$$\cos\left(\frac{\pi}{4} - x\right) \cos 2x + \sin x \sin 2x \sec x$$

$$= \cos x \sin 2x \sec x + \cos\left(\frac{\pi}{4} + x\right) \cos 2x$$

then the possible value of $\sec x$ is (IIT-2013)

A) 1

B) -1

C) $\sqrt{2}$

D) $-\sqrt{2}$

Ans: C

$$\text{Sol: } \left(\cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{\pi}{4} + x\right) \right) \cos 2x$$

$$= (\cos x \sin 2x - \sin x \sin 2x) \sec x$$

$$\sqrt{2} \sin x \cos 2x = (\cos x - \sin x) 2 \sin x \cos x \sec x$$

$$\sin x + \cos x = \sqrt{2}$$

$$x = \frac{\pi}{4} \Rightarrow \sec x = \sqrt{2}.$$

W.E-17: The arithmetic mean of the root of the

$$\text{equations } 4 \cos^3 x - 4 \cos^2 x - \cos(\pi + x) - 1 = 0$$

in $(0, 315)$ is equal to

A) 49π

B) 50π

C) 51π

D) 100π

Ans: C

$$\text{Sol: } (4 \cos^2 x + 1)(\cos x - 1) = 0$$

$$\cos x = 1$$

$$x = 2\pi, 4\pi, \dots, 100\pi$$

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$$100\pi < 315 < 101\pi$$

$$A.M = \frac{2(\pi + 2\pi + \dots + 50\pi)}{50} = 51\pi$$

W.E-18: If $2 \tan^2 x - 5 \sec x = 1$ for exactly seven

distinct roots in of $x \in \left[0, \frac{n\pi}{2}\right], n \in N$ then

the greatest value of n is

A) 6

B) 12

C) 13

D) 15

Ans: D

$$\text{Sol: } 2 \sec^2 x - 5 \sec x - 3 = 0$$

$$(\sec x - 3)(2 \sec x + 1) = 0$$

$$\sec x = 3 \text{ or } \sec x = -\frac{1}{2}$$

$$|\sec x| \geq 1 \Rightarrow \sec x = 3$$

which gives two values of x in each of

$(0, 2\pi], (2\pi, 4\pi], (4\pi, 6\pi]$ and one of value in

$$\left(6\pi, 6\pi + \frac{\pi}{2}\right) = \left(6\pi, 15\frac{\pi}{2}\right)$$

$$n = 15.$$

W.E-19: If

$$f(x) = \begin{vmatrix} \sin^2 \theta & \cos^2 \theta & x \\ \cos^2 \theta & x & \sin^2 \theta \\ x & \sin^2 \theta & \cos^2 \theta \end{vmatrix}, \theta \in \left(0, \frac{\pi}{2}\right)$$

then the roots of $f(x) = 0$ are

A) $\frac{1}{2}, -1$

B) $\frac{1}{2}, -1, 0$

C) $-\frac{1}{2}, 1, 0$

D) $-\frac{1}{2}, -1, 0$

Ans: A

$$\text{Sol: } C_1 \rightarrow C_1 + C_2 + C_3$$

$$f(x) = (1+x) \begin{vmatrix} 1 & \cos^2 \theta & x \\ 1 & x & \sin^2 \theta \\ 1 & \sin^2 \theta & \cos^2 \theta \end{vmatrix}$$

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$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$f(x) = (1+x) \begin{vmatrix} 1 & \cos^2 \theta & x \\ 0 & x - \cos^2 \theta & \sin^2 \theta - x \\ 0 & \sin^2 \theta - \cos^2 \theta & \cos^2 \theta - x \end{vmatrix}$$

$$= -\frac{1}{2}(1+x)[(x - \sin^2 \theta)^2 + (x - \cos^2 \theta)^2 + (\sin^2 \theta - \cos^2 \theta)^2]$$

$$f(x) = 0 \text{ if}$$

$$x = -1 \text{ or } x = \cos^2 \theta = \sin^2 \theta, \theta = \frac{\pi}{4}$$

$$\therefore x = -1 \text{ or } \frac{1}{2}.$$

W.E-20: Number of ordered pairs which satisfy the equation $x^2 + 2x \sin(xy) + 1 = 0$, where

$y \in [0, 2\pi]$, is

- A) 1 B) 2
C) 3 D) 0

Ans: B

Sol: Given $x^2 + 2x \sin(xy) + 1 = 0$

$$\Rightarrow [x + \sin(xy)]^2 + [1 - \sin^2(xy)] = 0$$

$$\Rightarrow x + \sin(xy) = 0 \text{ and } \sin^2(xy) = 1$$

$$\sin^2(xy) = 1 \text{ gives } \sin(xy) = 1 \text{ or } -1$$

$$\text{If } \sin(xy) = 1 \Rightarrow x = -1 \Rightarrow \sin(-y) = 1 \Rightarrow \sin y = -1,$$

Then the ordered pair is $(1, 3\pi/2)$

$$\text{if } \sin(xy) = -1 \Rightarrow x = 1 \Rightarrow \sin y = -1 \Rightarrow (-1, 3\pi/2)$$

Thus there are two ordered pairs.

LEVEL-V

SINGLE CORRECT CHOICE

1. The number of distinct real roots of the equation $\sin^3 x + \sin^2 x + \sin x - \sin x \cdot \sin 2x - \sin 2x - 2 \cos x = 0$

belonging to the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is

- (A) 0 (B) 1 (C) 2 (D) 3

2. The number of distinct real roots of the equation $\cos^8 x - \sin^8 x = 1, 0 \leq x \leq 2\pi$ is
(A) 4 (B) 8 (C) 3 (D) 6

3. The number of distinct real roots of the equation $\sqrt{\sin x} - \frac{1}{\sqrt{\sin x}} = \cos x$ is

(where $0 \leq x \leq 2\pi$)

- (A) 1 (B) 2
(C) 3 (D) more than 3

4. The number of distinct real roots of the equation $\sin \pi x = x^2 - x + \frac{5}{4}$ is

- (A) 0 (B) 1 (C) 2 (D) 4

5. If α, β are two distinct solutions lying between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ of the equation

$$2 \tan x + \sec x = 2 \text{ then } \tan \alpha + \tan \beta =$$

- (A) 0 (B) 1 (C) $\frac{4}{3}$ (D) $\frac{8}{3}$

6. In the range of $0 \leq x \leq 5\pi$, then equation $(3 + \cos x)^2 = 4 - 2 \sin^8 x$ has

- (A) exactly one solution
(B) exactly three solutions
(C) exactly five solution
(D) infinite solutions

7. The number of solutions of the pair equations $2 \sin^2 \theta - \cos 2\theta = 0$,

$$2 \cos^2 \theta - 3 \sin \theta = 0, \text{ in the interval } [0, 2\pi] \text{ is}$$

- A) zero B) one C) two D) four

8. If m and $n(>m)$ are positive integers, the number of solutions of the equation $n|\sin x| = m|\cos x|$ in $[0, 2\pi]$ is

- A) m B) n C) mn D) 4

9. If $[\sin x] + [\sqrt{2} \cos x] = -3, x \in [0, 2\pi]$ ($[\cdot]$ denotes the greatest integer function), then x belongs to

- A) $\left(\pi, \frac{5\pi}{4}\right)$ B) $\left[\pi, \frac{5\pi}{4}\right]$
 C) $\left(\frac{5\pi}{4}, 2\pi\right)$ D) $\left[\frac{5\pi}{4}, 2\pi\right]$

10. If $f(x) = \max\{\tan x, \cot x\}$. The number of roots of the equation

$$f(x) = \frac{1}{2 + \sqrt{3}} \text{ in } (0, 2\pi) \text{ is}$$

- A) 0 B) 2
 C) 4 D) infinite

11. Values of x and y satisfying the equation

$$\sin^7 y = |x^3 - x^2 - 9x + 9| + |x^3 - x^2 - 4x + 4| + \sec^2 2y + \cos^4 y$$

are

- A) $x = 1, y = n\pi, n \in I$
 B) $x = 1, y = 2n\pi + \frac{\pi}{2}, n \in I$
 C) $x = 1, y = 2n\pi, n \in I$
 D) none of the above

12. If $[y] = [\sin x]$ and $y = \cos x$ are two given equations, then the number of solutions is (where $[.]$ denotes the greatest integer function)

- A) 2 B) 3
 C) 4 D) infinitely many solutions

13. The number of points inside the curve $x^2 + y^2 \leq 4$ satisfying

$$\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y \text{ is.}$$

- A) 4 B) 6 C) 8 D) 10

ONE OR MORE THAN ONE CORRECT ANSWERS

14. Consider the equation

$$\sqrt{2} \tan^2 x - \sqrt{10} \tan x + \sqrt{2} = 0; 0 < x < \frac{\pi}{2} \text{ then}$$

TRIGONOMETRIC EQUATIONS

the true statements among the following are

(A) The number of distinct real roots of the equation is 2

(B) The sum of all the real roots of the equation is $\frac{\pi}{2}$

(C) The absolute value of the difference of the real roots of the equation is $\frac{\pi}{4}$

(D) The equation has no real roots

15. The statement

$$(3 \sin^4 x - 2 \cos^6 x + y - 2 \sin^6 x + 3 \cos^4 x)^2 = 9$$

is true for

(A) $x = \frac{5\pi}{13}$ and $y = 2$

(B) $x = \frac{7\pi}{16}$ and $y = -4$

(C) $x = \frac{\pi}{7}$ and $y = 1$

(D) $z = \frac{\pi}{5}$ and $y = -1$

16. The solution set of $|\sin x| \leq |\cos 2x|$ contains

(A) $\bigcup_{n \in I} \left[n\pi - \frac{\pi}{6}, n\pi + \frac{\pi}{6} \right]$

(B) $\bigcup_{n \in I} \left[n\pi + \frac{\pi}{2} \right]$

(C) $\bigcup_{n \in I} \left[n\pi - \frac{\pi}{8}, n\pi + \frac{\pi}{8} \right]$

(D) $\bigcup_{n \in I} \left[n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{4} \right]$

17. If $\sin \theta = \cos \phi$, then the possible values of

$$\frac{1}{\pi} \left(\theta \pm \phi - \frac{\pi}{2} \right) \text{ are [IIT 2008]}$$

- (A) 0 (B) 1 (C) 2 (D) 3

18. For the smallest positive values of 'x' and 'y' the equation

$$2(\sin x + \sin y) - 2 \cos(x - y) = 3 \text{ has a}$$

TRIGONOMETRIC EQUATIONS

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solution then which of the following is/are true

(A) $\sin\left(\frac{x+y}{2}\right) = 1$

(B) $\cos\left(\frac{x-y}{2}\right) = \frac{1}{2}$

(C) number of ordered pairs (x, y) is 2

(D) number of ordered pairs (x, y) is 3

19. Which of the following inequalities hold true in any ΔABC ?

(A) $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \leq \frac{1}{8}$

(B) $\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2} \leq \frac{3\sqrt{3}}{8}$

(C) $\sin^2\frac{A}{2} + \sin^2\frac{B}{2} + \sin^2\frac{C}{2} < \frac{3}{4}$

(D) $\cos^2\frac{A}{2} + \cos^2\frac{B}{2} + \cos^2\frac{C}{2} \leq \frac{9}{4}$

20. If $\frac{\cos(\alpha-3\theta)}{\cos^3\theta} = \frac{\sin(\alpha-3\theta)}{\sin^3\theta} = m$, then

(A) $\cos 2\alpha = \frac{2m^4 - 9m^2 + 8}{m^2}$

(B) $\cos \alpha = \frac{2-m^2}{m}$

(C) $\cos 2\alpha = \frac{2m^4 + 9m^2 + 8}{m^2}$

(D) $\cos \alpha = \frac{2+m^2}{m}$

21. The equation $\sin x = [1 + \sin x] + [1 - \cos x]$ has (where $[x]$ is the greatest integer less than or equal to x)

A) no solution in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B) no solution in $\left[\frac{\pi}{2}, \pi\right]$

C) no solution in $\left[\pi, \frac{3\pi}{2}\right]$

D) no solution for $x \in R$

22. The solution of the equation

$9\cos^2 x + \cos^2 2x + 1 = 6\cos^6 x \cos 2x + 6\cos^6 x - 2\cos 2x$ is/are

A) $x = n\pi + \frac{\pi}{2}, n \in I$

B) $x = n\pi + \cos^{-1}\left(\sqrt[4]{\frac{2}{3}}\right), n \in I$

C) $x = n\pi - \cos^{-1}\left(\sqrt[4]{\frac{2}{3}}\right), n \in I$

D) none of the above

23. $2\sin^2\left(\frac{\pi}{2}\cos^2 x\right) = 1 - \cos(\pi \sin 2x)$, If

A) $x = (2n+1)\frac{\pi}{2}, n \in I$

B) $\tan x = \frac{1}{2}, n \in I$

C) $\tan x = -\frac{1}{2}, n \in I$

D) $x = \frac{n\pi}{2}, n \in I$

24. If $\sin\left(\frac{6}{5}x\right) = 0$ and $\cos\left(\frac{x}{5}\right) = 0$ then

A) $x = (n-5)\pi$ B) $x = 6(n-1)\pi$

C) $x = 5(n-\frac{1}{2})\pi$ D) $x = 5(n+\frac{1}{2})\pi$

COMPREHENSION PASSAGES

(P) A real number θ is a root of the equation

$7\cos^2 x + 4\cos x - 1 = 0$.

If $p \cos^2 2\theta + q \cos 2\theta + r = 0$ where p, q, r are real constants then

25. p can be
(A) 16 (B) 49 (C) 64 (D) 81
26. q can be
(A) 49 (B) 54 (C) 38 (D) 20
27. r can be
(A) -7 (B) -16 (C) 14 (D) -5
- (P) If curve of $y = f(x)$ and $y = g(x)$ intersects at n different points $x = x_1, x_2, x_3, \dots$ then equation $f(x) = g(x)$ is said to have n solutions
28. Number of solutions of $|\cos x| = 2[x]$ is (where $[x]$ is integral part of x)
(A) 0 (B) 1 (C) 2 (D) infinite
29. The number of solutions of $\sin \pi x = |\log_e |x||$ is
(A) 0 (B) 6 (C) 4 (D) 8
30. Number of solutions of the equation $\sin^5 x - \cos^5 x = \frac{1}{\cos x} - \frac{1}{\sin x}$ ($\sin x \neq \cos x$) is
(A) 0 (B) 1 (C) 2 (D) infinite

Passage-3:

Consider the system of equations

$$\sin x \cos 2y = (a^2 - 1)^2 + 1, \cos x \sin 2y = a + 1.$$

31. The number of values of a for which the system has a solution is
(A) 1 (B) 2 (C) 3 (D) Infinite
32. The number of values of $x \in [0, 2\pi]$ when the system has solution for permissible values of ' a ' is
(A) 1 (B) 2 (C) 3 (D) 4

Passage-4:

Consider the cubic equation

$$x^3 - (1 + \cos \theta + \sin \theta)x^2 + (\cos \theta \sin \theta + \cos \theta + \sin \theta)x - \sin \theta \cos \theta = 0$$

whose roots are x_1, x_2 , and x_3 .

33. The value of $x_1^2 + x_2^2 + x_3^2$ equals
(A) 1 (B) 2
(C) $2 \cos \theta$ (D) $\sin \theta (\sin \theta + \cos \theta)$
34. Number of values of θ in $[0, 2\pi]$ for which at least two roots are equal
(A) 3 (B) 4 (C) 5 (D) 6
35. Greatest possible difference between two of the roots if $\theta \in [0, 2\pi]$ is
(A) 2 (B) 1 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

ASSERTION AND REASONING

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

36. Statement 1 : The equations $\cos 2\theta = 1$ and

$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = 1$ do not have the same solution set.

Statement 2 : The functions $f(\theta) = \cos 2\theta$

and $g(\theta) = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ are not equal.

37. Statement 1: The equation $\tan x = x$ has infinitely many roots.

Statement 2: The straight line $y = x$ intersects the graph of $y = \tan x$ at infinitely many points.

38. Consider two functions $f(x) = 1 + e^{\cot^2 x}$

and $g(x) = \sqrt{2|\sin x| - 1} + \frac{1 - \cos 2x}{1 + \sin^4 x}$

Statement-1: The solutions of the equation $f(x) = g(x)$ is given by

$$x = (2n + 1)\frac{\pi}{2} \quad \forall n \in \mathbb{I}.$$

TRIGONOMETRIC EQUATIONS

Statement-2: If $f(x) \geq k$ and $g(x) \leq k$ (where $k \in \mathbb{R}$) then solutions of the equation $f(x) = g(x)$ is the solution corresponding to the equation $f(x) = k$.

MATRIX MATCHING

39. Column - I

A) Number of distinct real roots of the

equation $x = \left(\frac{5\pi}{2}\right)^{\cos x}$

B) If 'a' is irrational then the number of real roots of the equation

$1 + \sin^2 ax = \cos x$

C) The number of real roots of the equation $4\sin 2x + \cos 2x = 5$ is

D) The number of distinct real roots of the equation

$4\cos^2 x + 2(\sqrt{3} + 1)\sin x - \sqrt{3} - 4 = 0$ in the

interval $\left[0, \frac{\pi}{2}\right]$ is

Column - II

P) 0

Q) 1

R) 2

S) 3

T) 4

40. Column I (Equation)

A. $x^3 + x^2 + 4x + 2\sin x = 0$ in $0 \leq x \leq 2\pi$

B. $\sin e^x \cos e^x = 2^{x-2} + 2^{-x-2}$

C. $\sin 2x + \cos 4x = 2$

D. $30|\sin x| = x$ in $0 \leq x \leq 2\pi$

Column II (Solutions)

P. 4

Q. 1

R. 2

S. 0

INTEGER TYPE QUESTIONS

41. The number of distinct real roots of the equation $\tan^2 2x + 2 \tan 2x \tan 3x - 1 = 0$ in

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the interval $\left[0, \frac{\pi}{2}\right]$ is

42. Number of positive roots of the equation $2\sin x = [x]$ is ($[x]$ represents integral part of x)

43. Number of solutions of

$\sin x + \sin y = \sin(x + y)$ and $|x| + |y| = 1$ is

44. Number of solutions of the equation

$\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} = 0$ in the

interval $\left(0, \frac{\pi}{4}\right)$ is

KEY - LEVEL-V

SINGLE CORRECT CHOICE

- | | | | |
|---------|---------|---------|---------|
| 1. (B) | 2. (C) | 3. (A) | 4. (B) |
| 5. (D) | 6. (B) | 7. (C) | 8. (D) |
| 9. (A) | 10. (A) | 11. (B) | 12. (D) |
| 13. (A) | | | |

ONE OR MORE THAN ONE CORRECT ANSWERS

- | | |
|---------------|---------------|
| 14. (A,B) | 15. (A,B) |
| 16. (ABC) | 17. (A,C) |
| 18. (A,B,C) | 19. (A,B,D) |
| 20. (A,B) | 21. (A,B,C,D) |
| 22. (A, B, C) | 23. (A, B, C) |
| 24. (C, D) | |

COMPREHENSION PASSAGES

- | | | | |
|---------|---------|---------|---------|
| 25. (B) | 26. (C) | 27. (A) | 28. (A) |
| 29. (B) | 30. (A) | 31. (A) | 32. (B) |
| 33. (B) | 34. (C) | 35. (A) | |

ASSERTION AND REASONING

36. (D) 37. (A) 38. (C)

MATRIX MATCHING

39. (A-Q, B-Q, C-P, D-R)

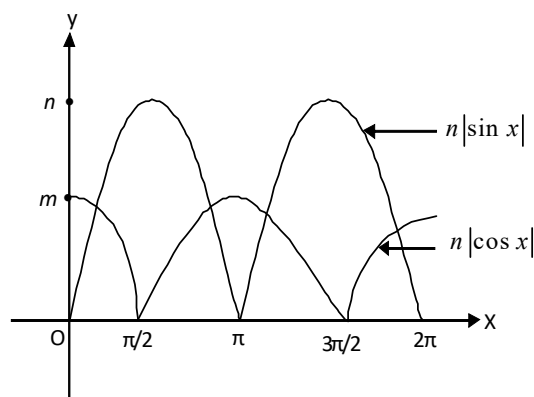
40. (A-Q, B-S, C-S, D-P)

INTEGER TYPE QUESTIONS

41. (3) 42. (0) 43. (6) 44. (6)

HINTS - LEVEL-V**SINGLE CORRECT CHOICE**

- $(\sin^2 x + \sin x + 1)(\sin x - 2\cos x) = 0$
 $\Rightarrow \tan x = 2$
- $\cos^8 x = 1 + \sin^8 x$
 $\text{L.H.S} \leq 1$ and $\text{R.H.S} \geq 1$
 $\therefore \text{L.H.S} = \text{R.H.S} = 1$
 $\therefore \sin x = 0$ and $\cos^2 x = 1$
- The equation clearly holds if $\sin x = 1$
- $\sin \pi x \leq 1$ and $x^2 - x + \frac{5}{4} \geq 1$
 $\therefore \sin \pi x = 1$ and $x = \frac{1}{2}$
- $\sec^2 x = (2 - 2\tan x)^2$
 $\Rightarrow 3\tan^2 x - 8\tan x + 3 = 0$
 $\tan \alpha, \tan \beta$ are the roots of this equation.
- $4 \leq \text{LHS} \leq 16$
 $2 \leq \text{RHS} \leq 4$
hence equality can occur at 4, which is possible
if $x = \pi, 3\pi, 5\pi \Rightarrow 3 \text{ solutions}$
- From the given equations $\sin^2 \theta = \frac{1}{4}$ and
 $\sin \theta = \frac{1}{2}$ which has two solutions in $[0, 2\pi]$
- Let $y = n|\sin x| = m|\cos x|$
The curve $y = n|\sin x|$
and $y = m|\cos x|$.
Intersect at 4 points in $[0, 2\pi]$.



- $[\sin x] + [\sqrt{2} \cos x] = -3$
 $\Rightarrow [\sin x] = -1$ and $[\sqrt{2} \cos x] = -2$
or $-1 \leq \sin x < 0$
and $-2 \leq \sqrt{2} \cos x < -1$
or $-1 \leq \sin x < 0$
and $-\sqrt{2} < \cos x < -\frac{1}{\sqrt{2}}$
or $-1 \leq \sin x < 0$
and $-1 \leq \cos x < -\frac{1}{\sqrt{2}}$
 $\therefore x \in (\pi, 2\pi)$ and $x \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$
 $\therefore x \in (\pi, 2\pi) \cap \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$
 $\therefore x \in \left(\pi, \frac{5\pi}{4}\right)$
- Least value of
 $\max\{\tan x, \cot x\}$ in $(0, 2\pi)$ is 1,
But $1 > \frac{1}{2 + \sqrt{3}}$, Hence
 $\max\{\tan x, \cot x\} = \frac{1}{2 + \sqrt{3}}$ has no solution.
-

TRIGONOMETRIC EQUATIONS

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$$\sin^7 y = |x^3 - x^2 - 9x + 9| + |x^3 - x^2 - 4x + 4| + \sec^2 2y + \cos^4 y$$

for $x = 1$

$$\sin^7 y = \sec^2 2y + \cos^4 y$$

$$\Rightarrow \sin^7 y \cos^2 2y = 1 + \cos^4 y \cos^2 2y$$

Since, LHS ≤ 1 .

and RHS ≥ 1

which is possible only when

$$\therefore \sin^7 y \cos^2 2y = 1$$

$$\Rightarrow \sin^7 y = 1 \text{ and } \cos^2 2y = 1$$

$$y = \frac{\pi}{2}$$

General value of y is $2n\pi + \frac{\pi}{2}$

Hence, $x = 1$ and $y = 2n\pi + \frac{\pi}{2}, n \in I$

12. $\therefore [y] = [\sin x] = -1, 0, 1$

$$\because -1 \leq \sin x \leq 1$$

$$\Rightarrow -1 \leq y < 0$$

$$\therefore [\sin x] = -1, 0, 1$$

$$[\sin x] = -1$$

$$\Rightarrow -1 \leq \sin x < 0$$

Then, $x \in \{(2n+1)\pi, (2n+2)\pi\}, n \in I$

and $[y] = 0$

$$\Rightarrow 0 \leq y < 1$$

$$[\sin x] = 0$$

$$\therefore 0 \leq \sin x < 1$$

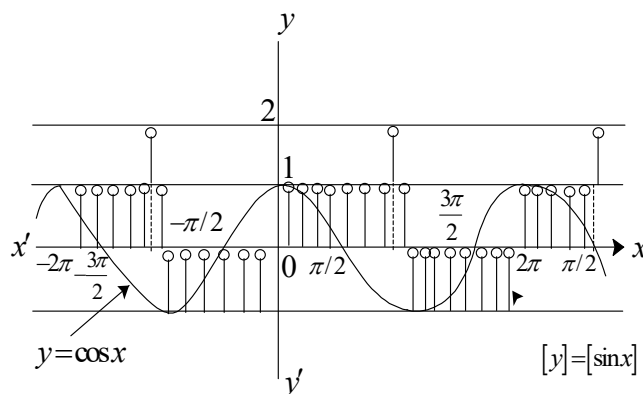
Then,

$$x \in [2n\pi, (2n+1)\pi] \sim \left\{2n\pi + \frac{\pi}{2}\right\}, n \in I$$

and $[y] = 1, \sin x = 1$

$$x = 2n\pi + \frac{\pi}{2}$$

$$1 \leq y < 2$$



Hence, $[y] = [\sin x]$ and $y = \cos x$ infinite points. Hence, number of solutions = infinite.

13. $LHS = (\tan^2 x - \cot^2 x^2) + 3 \geq 3$

$$RHS \leq 3$$

Thus, given equation have a solution, if

$$LHS = RHS = 3$$

$$\text{or } \tan^2 x = \cot^2 x \text{ and } \sin^2 y = 1$$

$$\therefore x = \pm\pi/4, \pm3\pi/4$$

$$\text{and } y = \pm\pi/2, \pm3\pi/2$$

Clearly, $x^2 + y^2 \leq 4$ is satisfied by

$$x = \pm\pi/4, y = \pm\pi/2$$

Only four pair of solutions are possible.

ONE OR MORE THAN ONE CORRECT ANSWERS

14. Solving, we get $\tan x_1 = \frac{\sqrt{5}+1}{2}$ and

$$\tan x_2 = \frac{\sqrt{5}-1}{2}$$

$$\therefore \tan(x_1 + x_2) \text{ is undefined}$$

$$\tan(x_1 - x_2) = \frac{1}{2}$$

15. $3(\cos^4 x + \sin^4 x) - 2(\cos^6 x + \sin^6 x) = 1$

$$\therefore (y+1)^2 = 9 \Rightarrow y = 2 \text{ or } -4$$

$$\begin{aligned}
 16. \quad |\cos 2x|^2 &\geq |\sin x|^2 \Rightarrow \cos^2 2x \geq \frac{1 - \cos 2x}{2} \\
 &\Rightarrow (\cos 2x + 1)(2 \cos 2x - 1) \geq 0 \\
 &\Rightarrow \text{either } \cos 2x = -1 \text{ or } \cos 2x \geq \frac{1}{2} \\
 &\Rightarrow \text{either} \\
 2x &\in 2n\pi + \pi \text{ or } \left[2n\pi - \frac{\pi}{3}, 2n\pi + \frac{\pi}{3}\right], n \in \mathbb{I}.
 \end{aligned}$$

$$17. \quad \sin \theta = \cos \phi \Rightarrow \cos\left(\frac{\pi}{2} - \theta\right) = \cos \phi$$

$$\frac{\pi}{2} - \theta = 2n\pi \pm \phi$$

$$\frac{1}{\pi}\left(\theta \pm \phi - \frac{\pi}{2}\right) = -2n$$

$$\Rightarrow 0 \text{ and } 2 \text{ are possible}$$

$$18. \quad 2(\sin x + \sin y) - 2 \cos(x - y) = 3$$

$$2\left[2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}\right] - 2\left[2 \cos^2 \frac{x-y}{2} - 1\right] = 3$$

$$\text{Let } \cos \frac{x-y}{2} = t$$

$$\left(4 \sin \frac{x+y}{2}\right)t - 4t^2 + 2 = 3$$

$$\Rightarrow 4t^2 - 4t \sin\left(\frac{x+y}{2}\right) + 1 = 0$$

$$\text{Since } t \text{ is real } D \geq 0$$

$$16 \sin^2 \frac{x+y}{2} - 16 \geq 0$$

$$\sin^2 \frac{x+y}{2} \geq 1$$

$$\text{Also } \sin^2 \frac{x+y}{2} \leq 1$$

$$\therefore \sin^2 \frac{x+y}{2} = 1$$

$$\Rightarrow \sin \frac{x+y}{2} = \pm 1$$

$$\text{For } \sin \frac{x+y}{2} = 1, 4t^2 - 4t + 1 = 0$$

$$\Rightarrow (2t-1)^2 = 0 \Rightarrow t = \frac{1}{2}$$

$$\Rightarrow \cos\left(\frac{x-y}{2}\right) = \frac{1}{2}$$

$$\frac{x+y}{2} = \frac{\pi}{2}, \frac{x-y}{2} = \frac{\pi}{3}$$

$$\Rightarrow x+y = \pi \text{ \& } x-y = \frac{2\pi}{3}$$

$$\Rightarrow 2x = \frac{5\pi}{3}, 2y = \frac{\pi}{3} \Rightarrow x = \frac{5\pi}{6}, y = \frac{\pi}{6}$$

For

$$\sin \frac{x+y}{2} = -1 \Rightarrow \frac{x+y}{2} = \frac{3\pi}{2} \Rightarrow x+y = 3\pi$$

$$x-y = \frac{2\pi}{3} \Rightarrow 2x = \frac{11\pi}{3} \text{ \& } 2y = \frac{7\pi}{3}$$

$$\Rightarrow x = \frac{11\pi}{6} \text{ \& } y = \frac{7\pi}{6} \quad \text{Ordered}$$

$$\text{pairs are } \left(\frac{5\pi}{6}, \frac{\pi}{6}\right) \text{ \& } \left(\frac{11\pi}{6}, \frac{7\pi}{6}\right)$$

$$19. \quad \text{Let } k = \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\Rightarrow 2k = \left(2 \sin \frac{A}{2} \sin \frac{B}{2}\right) \sin \frac{C}{2}$$

$$\Rightarrow 2k = \left(\cos \frac{A-B}{2} - \cos \frac{A+B}{2}\right) \sin \frac{C}{2}$$

$$\Rightarrow 2k = \cos \frac{A-B}{2} \sin \frac{C}{2} - \sin^2 \frac{C}{2}$$

$$\Rightarrow \sin^2 \frac{C}{2} - \sin \frac{C}{2} \cos \frac{A-B}{2} + 2k = 0$$

$$\therefore \sin \frac{C}{2} \text{ is real}$$

$$D \geq 0$$

$$\cos^2 \frac{A-B}{2} - 8k \geq 1$$

$$8k \leq \cos^2 \frac{A+B}{2} \leq 1$$

$$k \leq \frac{1}{8}$$

Maximum occurs for equilateral triangle

Similarly for other choice has occurs for equilateral triangle.

$$20. \quad \cos(\alpha - 3\theta) = m \cos^3 \theta$$

$$\sin(\alpha - 3\theta) = m \sin^3 \theta$$

$$\cos \alpha = \cos(\alpha - 3\theta + 3\theta)$$

$$= \cos(\alpha - 3\theta) \cos 3\theta - \sin(\alpha - 3\theta) \sin 3\theta$$

$$= m \cos^3 \theta (4 \cos^3 \theta - 3 \cos \theta) - m \sin^3 \theta (3 \sin \theta - 4 \sin^3 \theta)$$

$$\cos \alpha = m \left[4(\cos^6 \theta + \sin^6 \theta) - 3(\cos^4 \theta + \sin^4 \theta) \right]$$

$$= m \left[4(1 - 3 \sin^2 \theta \cos^2 \theta) - 3(1 - 2 \sin^2 \theta \cos^2 \theta) \right]$$

$$= m(1 - 6 \sin^2 \theta \cos^2 \theta)$$

$$\rightarrow (1)$$

Also

$$\cos^2(\alpha - 3\theta) + \sin^2(\alpha - 3\theta) = m^2(\cos^6 \theta + \sin^6 \theta)$$

$$\Rightarrow 1 = m^2(1 - 3 \sin^2 \theta \cos^2 \theta) \rightarrow (2)$$

$$\text{From (1) \& (2) } \cos \alpha = \frac{2 - m^2}{m}$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = \frac{2m^4 - 9m^2 + 8}{m^2}$$

$$21. \quad \therefore \sin x = [1 + \sin x] + [1 - \cos x]$$

$$\Rightarrow \sin x = 2 + [\sin x] + [-\cos x]$$

$$\text{At } x = -\frac{\pi}{2} \quad \sin x = 2 - 1$$

$$\therefore \sin x = 1 \quad -1 = 1 \text{ (impossible)}$$

$$\text{At } x = \frac{\pi}{2} \quad 1 = 2 + 1 - 0 \quad 1 = 3 \text{ (impossible)}$$

$$\text{At } x = \pi, 0 = 1 \quad \text{(impossible)}$$

$$\text{In } x \in \left(-\frac{\pi}{2}, 0\right), \sin x = 2 + (-1) + 0 = 1 \text{ (impossible)}$$

$$\text{In } x \in \left(0, \frac{\pi}{2}\right), \sin x = 2 + 0 - 1 = 1 \text{ (impossible)}$$

$$\text{In } x \in \left(\frac{\pi}{2}, \pi\right), \sin x = 2 + 0 + 0 = 2 \text{ (impossible)}$$

$$\text{In } x \in (\pi, 3\pi/2), \sin x = 2 - 1 + 0 = 1 \text{ (impossible)}$$

$$22. \quad \cos^2 x + \cos^2 2x + 1 - 6 \cos^6 x \cos 2x - 6 \cos^6 x + 2 \cos 2x = 0$$

$$\Rightarrow (3 \cos^6 x - 1 - \cos 2x)^2 = 0$$

$$\Rightarrow \cos^2 x (3 \cos^4 x - 2) = 0$$

$$\Rightarrow \cos x = 0$$

$$\Rightarrow x = n\pi + \frac{\pi}{2}, n \in I$$

$$\text{and } \cos^4 x = \frac{2}{3}$$

$$\Rightarrow \cos^2 x = \pm \sqrt{\frac{2}{3}}$$

$$\therefore \cos x = \pm \sqrt[4]{\left(\frac{2}{3}\right)}$$

$$\Rightarrow x = n\pi \pm \cos^{-1} \sqrt[4]{\left(\frac{2}{3}\right)}, n \in I$$

$$23. \quad 2 \sin^2 \left\{ \left(\frac{\pi}{2} \right) \cos^2 x \right\} = 1 - \cos(\pi \sin 2x)$$

$$2 \sin^2 \left\{ \left(\frac{\pi}{2} \right) \cos^2 x \right\} = 2 \sin^2 \left\{ \left(\frac{\pi}{2} \right) \sin 2x \right\}$$

$$\text{or if } \left(\frac{\pi}{2} \right) \cos^2 x = \pm \left(\frac{\pi}{2} \right) \sin 2x$$

or either $\cos x = 0$ or $\cos x = \pm 2 \sin x$

or if $x = (2n+1)\left(\frac{\pi}{2}\right), n \in I$

or $\tan x = \pm \frac{1}{2}$

24. A) is false since, $\cos(n-5)\frac{\pi}{5} \neq 0$ for all integers n .
 B) is false, since $\sin \frac{6}{5} \cdot 6(n-1)\pi \neq 0$ for all integers n .
 C) is correct, since

$$\sin \frac{6}{5}x = \sin(6n-3)\pi = 0. \text{ Also}$$

$$\cos \frac{x}{5} = \cos\left(n - \frac{1}{2}\right)\pi = \cos(2n-1)\frac{\pi}{2} = 0$$

D) is also correct since

$$\sin \frac{6}{5}x = \sin(6n+3)\pi = 0 \quad \text{Also}$$

$$\cos \frac{x}{5} = \cos\left(n + \frac{1}{2}\right)\pi$$

$$\cos(2n+1)\frac{\pi}{2} = 0.$$

COMPREHENSION PASSAGES

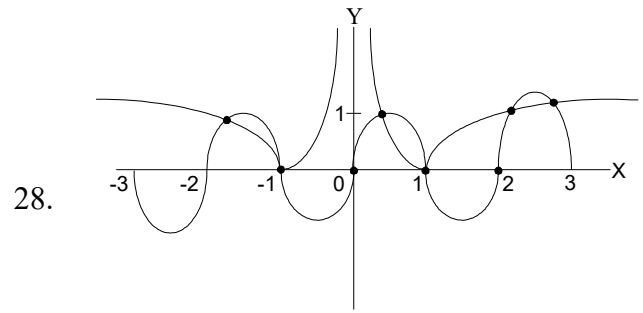
25 to 27

θ clearly satisfies $\left[(7\cos^2 x - 1) + 4\cos x\right]$

$$\left[(7\cos^2 x - 1) - 4\cos x\right] = 0$$

$$\Rightarrow (7\cos^2 \theta - 1)^2 - 16\cos^2 \theta = 0 \quad \text{Replacing}$$

$$2\cos^2 \theta = 1 + \cos 2\theta$$



29-30. $\sin^5 x - \cos^5 x = \frac{\sin x - \cos x}{\sin x \cos x}$

$$\Rightarrow \sin x \cos x \left[\frac{\sin^5 x - \cos^5 x}{\sin x - \cos x} \right] = 1$$

$$\sin 2x$$

$$\left[\sin^4 x + \sin^3 x \cos x + \sin^2 x \cos^2 x + \sin x \cos^3 x + \cos^4 x \right]$$

$$= 2$$

$$(\sin 2x - 2)^2 (\sin 2x + 2) = 0$$

$$\sin 2x = \pm 2$$

no solution

31-32

$$\sin x \cos 2y = (a^2 - 1)^2 + 1 \text{ and}$$

$$\cos x \sin 2y = a + 1 \rightarrow (1)$$

Since the LHS of the equations do not exceed 1, the given system may have solutions only for a 's such that

$$(a^2 - 1)^2 + 1 \leq 1 \text{ and } -1 \leq a + 1 \leq 1$$

$$(a^2 - 1)^2 + 1 \leq 1 \Rightarrow (a^2 - 1)^2 \leq 0 \Rightarrow a = \pm 1$$

But $a = 1$ does not satisfy (1)

$$\therefore a = -1$$

$$\sin x \cos 2y = 1$$

$$\cos x \sin 2y = 0$$

Adding & subtracting we get

$$\sin(x + 2y) = 1$$

$$\sin(x - 2y) = 1$$

$$x + 2y = 2n\pi + \frac{\pi}{2} \text{ and}$$

$$x - 2y = 2m\pi + \frac{\pi}{2} \quad n, m \in I$$

TRIGONOMETRIC EQUATIONS

Hence $x = (n+m)\pi + \frac{\pi}{4}; y = \frac{1}{2}(n-m)\pi, n, m \in I$

33-35

33.

$$x^3 - (1 + \cos\theta + \sin\theta)x^2 + (\cos\theta\sin\theta + \cos\theta + \sin\theta)x - \sin\theta\cos\theta = 0$$

Given cubic function is

$$f(x) = (x-1)(x-\cos\theta)(x-\sin\theta).$$

Therefore, roots are 1, $\sin\theta$, and $\cos\theta$.

Hence,

$$x_1^2 + x_2^2 + x_3^2 = 1 + \sin^2\theta + \cos^2\theta = 2$$

34. Now if $1 = \sin\theta \Rightarrow \theta = \frac{\pi}{2}$

if $1 = \cos\theta \Rightarrow \theta = 0, 2\pi$ and if

$$\sin\theta = \cos\theta \Rightarrow \tan\theta = 1$$

$$\therefore \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

$\therefore$ The number of values of θ in $[0, 2\pi]$ is 5.

35. Again the maximum possible difference between the two roots is 2

$$1 - \sin\theta = 2 \text{ when } \theta = 3\pi/2$$

$$\text{or } 1 - \cos\theta = 2 \text{ when } \theta = \pi$$

ASSERTION AND REASONING

36. $\cos 2\theta = \frac{1 - \tan^2\theta}{1 + \tan^2\theta}$ only when, $\cos\theta \neq 0$

37. Any root of the equation $\tan x = x$ is the abscissa of the point of intersection of the graphs of $y = x$ and $y = \tan x$.

38. L.H.S. $= 1 + e^{\cot^2 x} \geq 2$

$$\text{As } \sqrt{2|\sin x|} - 1 \leq 1$$

JEE ADVANCED - VOL - II

$$\text{and } \frac{1 - \cos 2x}{1 + \sin^4 x} = \frac{2 \sin^2 x}{1 + \sin^4 x} = \frac{2}{\frac{1}{\sin^2 x} + \sin^2 x}$$

≤ 1

$$\therefore \text{R.H.S.} = \sqrt{2|\sin x|} - 1 + \frac{1 - \cos 2x}{1 + \sin^4 x} \leq 2$$

Equation will satisfy if L.H.S. = R.H.S. = 2 which is possible when $\cot^2 x = 0$ and $|\sin x| = 1$

$$\Rightarrow x = (2n+1)\frac{\pi}{2}, n \in I$$

$\therefore$ Statement-1 is correct.

Statement-2 is not always correct because solution of the equation $f(x) = g(x)$ will be solutions corresponding to $f(x) = g(x) = k$ in the domain of $f(x)$ and $g(x)$ both.]

MATRIX MATCHING

39. (i) Draw the graphs of $y = \cos x$ and $y = \log x$

(ii) If 'a' is irrational then $x = 0$ is the only real root of the given equation.

(iii) $|4 \sin 2x + \cos 2x| \leq \sqrt{17}$

(iv) The given equation is a quadratic $\sin x$

$$\text{solving, } \sin x = \frac{\sqrt{3}}{2} \text{ or } \frac{1}{2}$$

40.

A. Here $x^3 + (x+2)^2 + 2 \sin x = 4$ Clearly, $x=0$ satisfies the equation.

$$\text{If } 0 < x \leq \pi, x^3 + (x+2)^2 + 2 \sin x > 4$$

If

$$\pi < x \leq 2\pi, x^3 + (x+2)^2 + 2 \sin x > 27 + 25 - 2$$

So $x = 0$ is the only solutions.

B. Here $\frac{1}{2} \sin(2e^x) = \frac{1}{4}(2^x + 2^{-x}) \geq \frac{1}{4} 2 = \frac{1}{2}$

$$\Rightarrow \sin(2e^x) \geq 1 \Rightarrow \sin(2e^x) = 1$$

But equality can hold when

$$2^x = 2^{-x} = 1, \text{ i.e., } x = 0$$

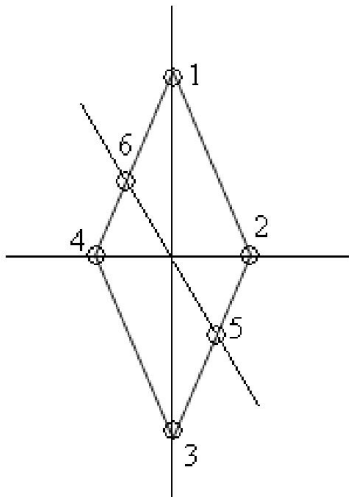
Then $\sin(2.e^0) = 1$, which is not true. Hence no solution.

- C. $\sin 2x + \cos 4x = 2$
 $\Rightarrow \sin 2x = 1, \cos 4x = 1$. Therefore,
 $1 - 2\sin^2 2x = 1$ or $1 - 2 = 1$ (absurd)

- D. The given solution is $|\sin x| x/30$. The solution of this equation are the point of intersections of the curves
 $y = |\sin x|$ and $y = x/30$.
 Since there are four points of intersection in $0 \leq x \leq 2\pi$, the equation has four solutions

INTEGER TYPE QUESTIONS

41. $\tan 2x(\tan 2x + \tan 3x) = 1 - \tan 2x \tan 3x$
 $\Rightarrow \tan 2x \tan 3x = 1 \Rightarrow \cos 7x = 0$
 42. Draw the graphs of $y = 2 \sin x$ and $y = [x]$
 43.



$$\sin x + \sin y = \sin(x + y)$$

$$2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x+y}{2}\right)$$

$$\sin\left(\frac{x+y}{2}\right) \left(\cos\frac{x-y}{2} - \cos\frac{x+y}{2} \right) = 0$$

$$\frac{x+y}{2} = 0 \quad (\text{or})$$

$$-2 \sin \frac{x}{2} \cdot \sin\left(\frac{-y}{2}\right) = 0$$

$$y = -x \quad x = 0 \quad (\text{or})$$

$$y = 0$$

Number of solutions : 6

$$44. \frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} = 0$$

$$\frac{1}{2} \left(\frac{\sin 2x^{3x-x}}{\cos x \cos x} + \frac{\sin 6x^{9x-3x}}{\cos 3x \cos 9x} + \frac{\sin 8x}{\cos 9x \cos 27x} \right) = 0$$

$$\frac{1}{2} ((\tan 3x - \tan x + \tan 9x - \tan 3x + \tan 27x - \tan 9x)) = 0$$

$$\tan 27x - \tan x = 0$$

$$\sin 27x \cdot \cos x - \sin x \cos 27x = 0$$

$$\sin(27x - x) = 0$$

$$\sin 26x = 0$$

$$26x = n\pi$$

$$x = \frac{n\pi}{26}$$

$$n = 1 \Rightarrow x = \frac{\pi}{26}, \frac{2\pi}{26}, \frac{3\pi}{26}, \frac{4\pi}{26}, \frac{5\pi}{26}, \frac{6\pi}{26}$$

Number of solutions : 6

LEVEL-VI

SINGLE CORRECT OPTION

1. $A = \{x \in R | \cos 2x + 3 \cos x + 2 = 0\}$ and

$B = \{x \in R | \sqrt{2}x - 5 < 1\}$. Then the number of elements in the set $A \cap B$ is

A) 0 B) 1 C) 2 D) 4

2. The number of distinct real values of the parameter 'a' for which the equation

$$\int_0^x \sin^2 \frac{t}{2} dt = a^2 x^2 - \left(\frac{3x-1}{2} \right) + \frac{1}{a^2}$$

possesses real roots belonging to the interval $[0, 4\pi]$

A) 0 B) 2 C) 4 D) infinite

3. The number of distinct real roots of the equation $\cos\left(\frac{\sqrt{2}+1}{2}x\right)\cos\left(\frac{\sqrt{2}-1}{2}x\right)=1$ is
 A) Infinite B) 1 C) 0 D) 4
4. The variable 'x' satisfying the equation $|\sin x \cos x| + \sqrt{2 + \tan^2 x + \cot^2 x} = \sqrt{3}$, belongs to the interval
 (A) $\left[0, \frac{\pi}{3}\right]$ (B) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$
 (C) $\left[\frac{3\pi}{4}, \pi\right)$ (D) non existent
5. The number of points P(x, y) lying inside or on the circle $x^2 + y^2 = 9$ and satisfying the equation $\tan^4 x + \cot^4 x + 2 = 4 \sin^2 y$, is
 (A) 2 (B) 4 (C) 8 (D) Infinite
6. The equation $2x = (2n + 1)\pi(1 - \cos x)$, (where n is a positive integer)
 (A) has infinitely many real roots
 (B) has exactly one real root
 (C) has exactly $2n + 2$ real roots
 (D) has exactly $2n + 3$ real roots
7. The number of solutions of the equation $16(\sin^5 x + \cos^5 x) = 11(\sin x + \cos x)$ in the interval $[0, 2\pi]$ is
 (A) 6 (B) 7 (C) 8 (D) 9
8. The number of non similar isosceles possible triangle's such that $\tan A + \tan B + \tan C = 100$ is
 A) 1 B) 2 C) 3 D) 4
9. The number of solutions of $[\sin x + \cos x] = 3 + [-\sin x] + [-\cos x]$ (where $[\cdot]$ denotes the greatest integer function), $x \in [0, 2\pi]$ is
 A) 0 B) 4 C) infinite D) 1
10. The equation $\cos^8 x + b \cos^4 x + 1 = 0$ will have a solution if b belongs to
 A) $(-\infty, 2]$ B) $[2, \infty)$
 C) $(-\infty, -2]$ D) $[-2, 0]$
11. The number of values of y in $[-2\pi, 2\pi]$ satisfying the equation $|\sin 2x| + |\cos 2x| = |\sin y|$ is
 A) 3 B) 4 C) 5 D) 6

12. If both the distinct roots of the equation $|\sin x|^2 + |\sin x| + b = 0$ in $[0, \pi]$ are real, then the values of b are
 A) $[-2, 0]$ B) $(-2, 0)$
 C) $[-2, 0)$ D) $(0, 0)$
13. The total number of solutions of $\sin\{x\} = \cos\{x\}$ (where $\{ \cdot \}$ denotes the fractional part) in $[0, 2\pi]$ is equal to
 A) 5 B) 6 C) 8 D) 9
14. The number of integral pairs (x, y) that satisfy the following two equations
$$\begin{cases} \cos(xy) = x \\ \tan(xy) = y \end{cases}$$
 is
 A) 1 B) 2 C) 4 D) 6
15. Number of solutions of the equation $\cos(\theta) \cos(\pi\theta) = 1$ is
 A) 0 B) 2 C) 1 D) infinite
16. Number of solutions of the equations $y = \frac{1}{3} \left[\sin x + \left[\sin x + \left[\sin x \right] \right] \right]$ and $[y + [y]] = 2 \cos x$, (where $[\cdot]$ denotes the greatest integer function) is
 A) 0 B) 1
 C) 2 D) infinite
17. The value of a for which the equation $4 \cos^2 \{ \pi(a + x) \} + a^2 - 4a = 0$ has a real solution, is
 A) 0 B) 1 C) 2 D) 3
18. The equation $\tan|x| = \tan x, x \in [-2\pi, 2\pi]$
 A) has exactly four solutions
 B) has exactly eight solutions
 C) infinite solutions
 D) has the set $\left\{-\frac{5\pi}{2}, -2\pi\right\}$ as solution
19. The greatest possible value of the expression

$$\tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$$

on the interval $[-5\pi/12, -\pi/3]$ is

- (A) $\frac{12}{5}\sqrt{2}$ (B) $\frac{11}{6}\sqrt{2}$
(C) $\frac{12}{5}\sqrt{3}$ (D) $\frac{11}{6}\sqrt{3}$

ONE OR MORE CORRECT ANSWERS

20. Consider the equation $\sqrt{\sin x} - \sqrt{\cos x} = a$ where $a \in R$ is a parameter. Then the true statements among the following are

- A) The equation has solution iff $|a| \leq 1$.
B) If $|a| \leq 1$ then the equation has no solution
C) $|a| \leq \frac{1}{\sqrt{3}}$ then the equation possess infinitely many solutions.
D) The equation has no solutions if $a \leq 0$.

21. Consider the equation

$$\cot x + \log_{\frac{\pi}{4}} x = 2, 0 < x < \pi \text{ then the true}$$

statements among the following are

- A) The equation has no real root
B) The equation has two distinct real roots.
C) The equation has a unique root.
D) A root of the equation lies between 0 and 1

22. Consider the equation

$$\int_0^x \cos(t + x^2) dt = \sin x, 2 \leq x \leq 3 \text{ then}$$

- A) The equation has no real root.
B) The equation has two distinct real roots.
C) The equation has exactly one real root.
D) $\frac{\sqrt{1+8\pi}-1}{2}$ is a root of the equation.

23. Consider the equation

$$-2\sqrt{3}\pi \sin x = |x + \pi| + |x - 2\pi| \text{ then the true}$$

TRIGONOMETRIC EQUATIONS

Statements among the following are.

- A) The equation has no real root.
B) The equation has exactly four distinct real roots.
C) All the roots of the equation are irrational.
D) All the roots of the equation lie between $-\pi$ and 2π .

24. Let $\theta, \phi \in [0, 2\pi]$ be such that

$$2 \cos \theta (1 - \sin \phi) = \sin^2 \theta \left(\tan \frac{\theta}{2} + \cot \frac{\theta}{2} \right) \cos \phi - 1,$$

$$\tan(2\pi - \theta) > 0 \text{ and } -1 < \sin \theta < -\frac{\sqrt{3}}{2}. \text{ Then}$$

ϕ cannot satisfy [IIT 2012]

- (A) $0 < \phi < \frac{\pi}{2}$ (B) $\frac{\pi}{2} < \phi < \frac{4\pi}{3}$
(C) $\frac{4\pi}{3} < \phi < \frac{3\pi}{2}$ (D) $\frac{3\pi}{2} < \phi < 2\pi$

25. The trigonometric equation $\sin 2x + \sin 3x + \sin 4x + \dots + \sin nx = n-i$ (n is a natural number greater than 2)

- (A) has unique solution for any n
(B) has no solution for any n
(C) has infinite solutions
(D) has no solution in $[2, n\pi]$

26. If $f(x) = [1 + \sin x] + \left[2 + \sin \frac{x}{2}\right] +$

$$\left[3 + \sin \frac{x}{3}\right] + \dots + \left[n + \sin \frac{x}{n}\right] \quad (0 < x < \pi)$$

A) min. value is $\frac{n(n+1)}{2}$

B) max. value is $\frac{n(n+1)}{2} + 1$

C) Range is $\left[\frac{n(n+1)}{2}, \frac{n(n+1)}{2} + 1\right]$

D) minimum value is $\frac{n(n+1)}{2} + 1$

- 27.

$$\sqrt{(\cos 2x)} + \sqrt{(1 + \sin 2x)} = 2\sqrt{(\sin x + \cos x)}, \text{ If}$$

A) $\sin x + \cos x = 0$

B) $x = 2n\pi, n \in I$

C) $x = n\pi - \frac{\pi}{4}, n \in I$

D) $x = 2n\pi \pm \cos^{-1}\left(-\frac{1}{5}\right), n \in I$

28. The inequation

$$2\sin^2\left(x - \frac{\pi}{3}\right) - 5\sin\left(x - \frac{\pi}{3}\right) + 2 > 0$$

A) is satisfied in $\left(-\frac{5\pi}{6}, \frac{5\pi}{6}\right)$

B) is satisfied in $\left(-\frac{5\pi}{6}, \frac{\pi}{2}\right)$

C) is satisfied in $\left(\frac{(12n-5)\pi}{6}, \frac{(4n+1)\pi}{2}\right)$

D) is satisfied for all x for which

$$\sin\left(x - \frac{\pi}{2}\right) < \frac{1}{2}$$

29. The equation $1 + \cos^2 ax = \sin x$, where a is a parameter

A) has a solution if a is not rational

B) does not have a solution if a is not rational

C) can have $a < \frac{\pi^2}{2}$ as a solution for some 'a'

D) can not have $(2009)\pi/2$ as a solution for any 'a'

30. The equation

$$\sin^2 x + \sin x - a = 0 (0 \leq x < 2\pi)$$

A) has solutions for every $a \geq -\frac{1}{4}$

B) has two solutions for $a = -\frac{1}{4}$

C) has four solutions for $-\frac{1}{4} < a < 0$

D) has two solutions for $-\frac{1}{4} < a < 0$

(P) Consider the equation

$$\tan^2\left(\cos\sqrt{4\pi^2 - x^2}\right)$$

$$-4a \tan\left(\cos\sqrt{4\pi^2 - x^2}\right) + 2 + 2a = 0, \text{ 'a' ,}$$

being a parameter. (Given $\tan 1 = 1.56$)

31. If a = 1 then the number of distinct real roots of the equation is

A) 0 B) 1 C) 2 D) Infinite

32. If $a = -\frac{1}{2}$ then the number of distinct real roots of the equation is

A) 0 B) 2 C) 3 D) 4

Passage-2:

Consider the equation

$$\sec \theta + \csc \theta = a, \theta \in (0, 2\pi) - \left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}$$

33. If the equation has four real roots, then

A) $|a| \geq 2\sqrt{2}$ B) $|a| < 2\sqrt{2}$

C) $|a| \geq -2\sqrt{2}$ D) $|a| > 2\sqrt{2}$

34. If the equation has exactly two real roots, then

A) $|a| \geq 2\sqrt{2}$ B) $a < 2\sqrt{2}$

C) $|a| < 2\sqrt{2}$ D) None

35. If the equation has exactly three real solutions then

A) $|a| \geq 4\sqrt{2}$

B) $a < 2\sqrt{2}$

C) $|a| < 2\sqrt{2}$

D) $a = 2\sqrt{2}$ or $a = -2\sqrt{2}$

Passage-3:

Suppose equation is $f(x) - g(x) = 0$ or

$f(x) = g(x) = y$ say, then draw the graphs

of $y = f(x)$ and $y = g(x)$. If graphs of

$y = f(x)$ and $y = g(x)$ cuts at one, two,

three,, no points, then number of solutions are one, two, three,, zero respectively.

On the basis of above information, answer the following questions:

36. The number of solutions of $\sin x = \frac{|x|}{10}$ is
 A) 4 B) 6 C) 8 D) none of these
37. Total number of solutions of the equation $3x + 2 \tan x = \frac{5\pi}{2}$ in $x \in [0, 2\pi]$ is equal to
 A) 1 B) 2 C) 3 D) 4
38. If $1 - \sin x = \frac{\sqrt{3}}{2} \left| x - \frac{\pi}{2} \right| + a$ has no solution when $a \in R^+$, then
 A) $a \in R^+$ B) $a > \frac{3}{2} - \frac{\pi}{\sqrt{3}}$
 C) $a \in \left(0, \frac{3}{2} + \frac{\pi}{\sqrt{3}} \right)$ D) $a \in \left(\frac{3}{2}, \frac{3}{2} + \frac{\pi}{\sqrt{3}} \right)$

MATRIX MATCHING

39. **Column - I**
 A) The set of all real values of parameter 'p' for which the equation $\sqrt{p} \cos x - 2 \sin x = \sqrt{2} + \sqrt{2-p}$ possesses at least one real root is
 B) The set of all real values of parameter 'a' for which the equation $\cos 2x + a \sin x = 2a - 7$ possesses at least one real root is
 C) The set of all real values of parameter 'a' for which the system $a \sin \frac{x}{2} = \sin x + \sin \frac{3x}{2}, \cos x \neq 1$ possesses at least one real root is
 D) The set of all real values of parameter 'a' for which the equation $\sin^4 x + \cos^4 x + \sin 2x + a = 0$ possesses at least one real root is
- Column-II**

P) $\left[-\frac{3}{2}, \frac{1}{2} \right]$

Q) $\left[-\frac{5}{4}, 5 \right]$

R) $[2, 6]$

S) $[\sqrt{5} - 1, 2]$

T) $[2, \sqrt{5} + 1]$

40. Observe the following columns:

Column-I

A) If α, β are the solutions of $\sin x = -\frac{1}{2}$ in $[0, 2\pi]$ and α, γ are solutions of

$$\cos x = \frac{-\sqrt{3}}{2} \text{ in } [0, 2\pi] \text{ then}$$

B) If α, β are the solutions of $\cot x = -\sqrt{3}$ in $[0, 2\pi]$ and α, γ are the solutions of $\operatorname{cosec} x = -2$ in $[0, 2\pi]$, then

C) If α, β are the solutions of $\sin x = \frac{1}{2}$ in $[0, 2\pi]$ and α, γ are the solutions

$$\text{of } \tan x = \frac{1}{\sqrt{3}} \text{ in } [0, 2\pi], \text{ then}$$

Column-II

P) $\alpha - \beta = \pi$

Q) $\beta - \gamma = \pi$

R) $\alpha - \gamma = \pi$

S) $\alpha + \beta = 3\pi$

T) $\beta + \gamma = 2\pi$

INTEGER ANSWER TYPE

41. The number of distinct real roots of

TRIGONOMETRIC EQUATIONS

$$\tan\left(\frac{\pi}{2}\cos x\right) = \cot(\pi\sin x), |x| < \frac{\pi}{2} \text{ is}$$

42. The number of distinct real roots of the

$$\text{equation } \tan\frac{2\pi x}{x^2+x+1} = -\sqrt{3} \text{ is}$$

43. If $a \in \mathbb{N}$ and $1 \leq a \leq 100$ then the number of distinct values of parameter 'a' for which the equation

$$\int_0^x (t^2 - 8t + 39) dx = x \sin \frac{\pi a}{x} \text{ possesses a}$$

real root is

44. Number of ordered pairs of real numbers which are solutions of the inequality

$$\sqrt{\log_x(\pi - \sqrt{y})} + 2 \cos(3\pi \cos \sqrt{y})$$

$$+ \sqrt{\log_{\pi - \sqrt{y}} x} \leq 0 \text{ is}$$

45. The number of distinct real roots of the equation $\sin 3x + \cos 2x + 2 = 0$ on the interval $[0, 2\pi]$ is

46. Let x be in radians with $0 < x < \frac{\pi}{2}$,

$$\sin(2\sin x) = \cos(2\cos x) \text{ if } \tan x + \cot x \text{ can}$$

be written as $\frac{a}{\pi^c - b}$ where $a, b, c \in \mathbb{N}$ then

the value of $\left(\frac{a+b+c}{25}\right)$ is

47. Number of real values of x , satisfying the equation $[x]^2 - 5[x] + 6 - \sin x = 0$ ($[.]$ denoting the greatest integer function) is

48. The number of solutions of the equation $|\cos |x|| - x = |x| \quad \forall x \in [-2\pi, 2\pi]$ is

49. The number of all possible values of θ where $0 < \theta < \pi$ for which the system of equations $(y+z)\cos 3\theta = (xyz)\sin 3\theta$

JEE ADVANCED - VOL - II

$$x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$$

$$(xyz)\sin 3\theta = (y+2z)\cos 3\theta + y \sin 3\theta$$

have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is
[IIT 2010]

50. The number of values of θ in the interval

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ such that } \theta \neq \frac{n\pi}{5}, \text{ for } n = 0, \pm 1,$$

± 2 and $\tan \theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$ is
[IIT 2010]

LEVEL - VI - KEY

SINGLE CORRECT OPITION

- | | | | |
|---------|---------|---------|---------|
| 1. (C) | 2. (C) | 3. (B) | 4. (D) |
| 5. (C) | 6. (C) | 7. (A) | 8. (B) |
| 9. (A) | 10. (C) | 11. (B) | 12. (B) |
| 13. (B) | 14. (A) | 15. (C) | 16. (A) |
| 17. (C) | 18. (C) | 19. (D) | |

ONE OR MORE CORRECT ANSWERS

- | | | |
|-------------|-------------|-----------|
| 20. (A,C) | 21. (C,D) | 22. (B,D) |
| 23. (B,C,D) | 24. (A,C,D) | |
| 25. (B,D) | 26. (A,B) | |
| 27. (A,B,C) | 28. (B,C,D) | |
| 29. (B,C) | 30. (B,C) | |

COMPREHENSION PASSAGES

- | | | | |
|---------|---------|---------|---------|
| 31. (A) | 32. (D) | 33. (D) | 34. (C) |
| 35. (D) | 36. (B) | 37. (C) | 38. (B) |

MATRIX MATCHING

39. (A-S, B-R, C-Q, D-P)
40. (A-Q, S, B-P, T, C-R, S, T)

INTEGER ANSWER TYPE

- | | | | |
|---------|---------|---------|---------|
| 41. (1) | 42. (5) | 43. (8) | 44. (3) |
| 45. (1) | 46. (2) | 47. (1) | 48. (3) |
| 49. (3) | 50. (3) | | |

HINTS LEVEL - VI

SINGLE CORRECT OPTION

1. $\cos x = -1$ or $-\frac{1}{2}$ and $2\sqrt{2} < x < 3\sqrt{2}$
 2. given equations is

$$\sin x = -1 - 2\left(ax - \frac{1}{a}\right)^2$$

$$L.H.S \geq -1 \text{ and } R.H.S \leq -1$$

$$\therefore \sin \frac{1}{a^2} = -1$$

3. The given equation can be written as
 $\cos x \sqrt{2} + \cos x = 2$

$$\therefore x\sqrt{2} = 2K\pi \text{ and } x = 2l\pi, K, l \in Z$$

clearly $x = 0$ is the only root.

4. $|\sin x \cos x| + |\tan x + \cot x| = \sqrt{3}$

$$\Rightarrow |\sin x \cos x| + \frac{1}{|\sin x \cos x|} = \sqrt{3}$$

$$\text{but } |\sin x \cos x| + \frac{1}{|\sin x \cos x|} \geq 2$$

hence, no solution

5. We have $\tan^4 x + \cot^4 x + 2 = 4\sin^2 y$

$$\Rightarrow \underbrace{(\tan^2 x - \cot^2 x)^2}_{\geq 4} + 4 = \underbrace{4\sin^2 y}_{\leq 4}$$

$$\therefore \tan^2 x = 1 \text{ and } \sin^2 y = 1 \Rightarrow \tan x = \pm 1 \text{ and } \sin y = \pm 1$$

$$\text{But } -3 \leq x \leq 3 \text{ and } -3 \leq y \leq 3$$

$$\therefore \text{Acceptable values of } x \text{ are } \pm \frac{\pi}{4} \text{ and } \pm \frac{3\pi}{4}$$

$$\text{and acceptable values of } y \text{ are } \pm \frac{\pi}{2}$$

Hence the number of points $P(x, y)$ are 8 **Ans.]**

6. $\sin^2\left(\frac{x}{2}\right) = \frac{x}{(2n+1)\pi}$

the graph of $\sin^2\left(\frac{x}{2}\right)$ will be above the x -axis and will be meeting the x -axis at $0, 2\pi, 4\pi, \dots$ etc.

TRIGONOMETRIC EQUATIONS

It will attain maximum values at odd multiples of π i.e. $\pi, 3\pi, \dots (2n+1)\pi$. The last point after

which graph of $Y = \frac{x}{(2n+1)\pi}$ will stop cutting

will be $(2n+1)\pi$.

$$\text{Total intersection} = 2(n+1)$$

$$7. 16(\sin^5 x + \cos^5 x) - 11(\sin x + \cos x) = 0$$

$$\Rightarrow (\sin x + \cos x)$$

$$\{16(\sin^4 x - \sin^3 x \cos x + \sin^2 x \cos^2 x - \sin x \cos^3 x + \cos^4 x) - 11\} = 0$$

$$\Rightarrow (\sin x + \cos x) \{16(1 - \sin^2 x \cos^2 x - \sin x \cos x) - 11\} = 0$$

$$\Rightarrow (\sin x + \cos x) (4\sin x \cos x - 1) (4\sin x \cos x + 5) = 0$$

As $4\sin x \cos x + 5 \neq 0$, we have

$$\sin x + \cos x = 0, 4\sin x \cos x - 1 = 0$$

The required values are

$$\pi/12, 5\pi/12, 9\pi/12, 13\pi/12,$$

$$17\pi/12, 21\pi/12, -$$

they are 6 solutions on $[0, 2\pi]$

8. $\tan A + \tan B + \tan C = 100$

Triangle ABC is isosceles triangle

Let $\angle A = \angle C$, then $\angle B = \pi - 2C$

$$\therefore 2\tan A - \tan 2A = 100$$

$$\tan^3 A - 50\tan^2 A + 50 = 0$$

$$x = \tan A$$

$$f(x) = x^3 - 50x^2 + 50$$

$$f'(x) = 3x^2 - 100x = 0$$

$$\text{Critical points } x = 0 \text{ and } x = \frac{100}{3}$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

$$\text{At } x = 0, f(x) = 50$$

$$\text{At } x = \frac{100}{3}, f(x) = \text{negative}$$

$\tan A$ takes one negative value which is not possible

($\because A = C$ and A acute)

$\Rightarrow \tan A$ has two distinct positive values

$\Rightarrow$ number of isosceles triangle = 2.

9. $[\sin x + \cos x] = 3 + [-\sin x] + [-\cos x]$

Maximum value of left-hand side is 1 and minimum of right hand side is also 1

$$\Rightarrow [\sin x + \cos x] = 3 + [-\sin x] + [-\cos x] = 1 \Rightarrow x \in \pi \pm \frac{\pi}{4}$$

$$\Rightarrow [\sin x + \cos x] = 1, [-\sin x] = -1, [-\cos x] = -1$$

which is not possible

$$10. \cos^8 x + b \cos^4 x + 1 = 0$$

$$\Rightarrow b = -\left(\cos^4 x + \frac{1}{\cos^4 x}\right) \leq -2 \forall x \in \mathbb{R}$$

$$\Rightarrow b \in (-\infty, -2]$$

$$11. \text{ Here } 1 \leq |\sin 2x| + |\cos 2x| \leq \sqrt{2} \text{ and } |\sin y| \leq 1$$

so solution is possible only when $|\sin y| = 1$

$$\Rightarrow \sin y = \pm 1 \Rightarrow y = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}$$

$$12. \text{ Given that } |\sin x|^2 + |\sin x| + b = 0$$

$$|\sin x| = \frac{-1 \pm \sqrt{1-4b}}{2}$$

$$\Rightarrow 0 \leq \frac{-1 \pm \sqrt{1-4b}}{2} < 1 \Rightarrow -2 < b < 0$$

$$13. \sin\{x\} = \cos\{x\}$$

$$\tan\{x\} = 1$$

$$\tan\left(\frac{\pi}{4}\right) = 1, \tan 1$$

Graphs of $y = \tan\{x\}$ and $y = 1$ meet exactly six times in $[0, 2\pi]$.

$$14. \frac{\sin(xy)}{\cos(xy)} = y$$

$$\Rightarrow \sin(xy) = xy$$

$$\Rightarrow xy = 0$$

$$\Rightarrow x = 0 \text{ or } y = 0$$

But $x = 0$ is not possible

$$\therefore y = 0 \text{ and } x = 1, \text{ i.e., } (1, 0)$$

$$15. \cos(\theta) \cos(\pi\theta) = 1$$

$$\Rightarrow \cos(\theta) = 1 \text{ and } \cos(\pi\theta) = 1 \quad \dots\dots (i)$$

$$\text{or } \cos(\theta) = -1 \text{ and } \cos(\pi\theta) = -1 \quad \dots\dots (ii)$$

Both the conditions cannot be satisfied

Therefore, $\theta = 0$ is only solution

$$16. y = \frac{1}{3} [\sin x + [\sin x + [\sin x]]]$$

$$= \frac{1}{3} ([\sin x] + [\sin x] + [\sin x])$$

$$= [\sin x]$$

$$\text{and } [y + [y]] = 2 \cos x$$

$$\Rightarrow 2[y] = (2 \cos x)$$

$$\Rightarrow [y] = \cos x$$

Case I: when $-1 \leq \sin x < 0$

$$\Rightarrow \therefore [\sin x] = -1$$

$$\text{Then, } y = -1$$

$$\text{ie, } [-1] = \cos x$$

$$\Rightarrow \cos x = -1$$

$$\Rightarrow \sin x = 0 \text{ (impossible)}$$

Case II: when $0 \leq \sin x < 1$

$$\therefore [\sin x] = 0$$

$$\text{Then, } y = 0$$

$$\text{ie, } 0 = \cos x$$

$$\Rightarrow \sin x = 1 \quad \text{(impossible)}$$

Case III: $\sin x = 1$

$$\therefore y = 1$$

$$\text{ie, } \cos x = 1$$

$$\Rightarrow \sin x = 0 \quad \text{(impossible)}$$

Hence, no solution

ie, no. of solutions is zero.

$$17. 4 \cos^2 \left\{ \pi(a+x) \right\} + a^2 - 4a = 0$$

$$\Rightarrow 4 \cos^2 \left\{ \pi(a+x) \right\} = 4a - a^2$$

$$\Rightarrow \sin^2 \left\{ \pi(a+x) \right\} = \frac{4}{4a - a^2}$$

For solution to exist

$$0 \leq \frac{4}{4a - a^2} \leq 1$$

$$\text{Now, } \frac{4}{4a - a^2} \geq 0$$

$$\Rightarrow 4a - a^2 > 0$$

$$\Rightarrow 0 < a < 4$$

.....(i)

$$\text{and } \frac{4}{4a - a^2} \leq 1$$

$$\Rightarrow \frac{a^2 - 4a + 4}{4a - a^2} \leq 0$$

$$\Rightarrow \frac{(a-2)^2}{4a - a^2} \leq 0$$

$$\Rightarrow \frac{(a-2)^2}{a(a-4)} \geq 0$$

$$\therefore a \in (-\infty, 0) \cup (4, \infty) \cup \{2\}$$

.....(ii)

From Eqs. (i) and (ii), $a = 2$.

18. If $x \in [-2\pi, 0]$ then the equation is
 $\tan|x| = \tan x$
 which is satisfied by all x in $[-2\pi, 0]$

Except $\frac{3\pi}{2}, \frac{\pi}{2}$

$\Rightarrow$ (c) is true and (a), (b) are false.

If $x \in \left(\frac{5\pi}{2}, 2\pi\right)$ then $|x| \in \left(2\pi, \frac{5\pi}{2}\right)$

$$\Rightarrow \tan|x| > 0, \text{ but } \tan x < 0$$

$\Rightarrow$ (d) is false

19. Let $u = -x - \pi/6$ then $u \in [\pi/6, \pi/4]$

and then $2u \in [\pi/3, \pi/2]$

$$\tan(x + 2\pi/3) = -\cot(x + \pi/6) = \cot u$$

now

$$\tan(x + 2\pi/3) - \tan(x + \pi/6) + \cos(x + \pi/6)$$

$$= \cot u + \tan u + \cos u$$

$$= \frac{2}{\sin 2u} + \cos u$$

both $\frac{2}{\sin 2u}$ and $\cos u$ monotonic

decreasing on $[\pi/6, \pi/4]$ and thus
 the greatest value occurs at

$$u = \pi/6$$

i.e

$$\frac{2}{\sin \pi/3} + \cos \pi/6 = \frac{4}{\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{11}{2\sqrt{3}} = \frac{11\sqrt{3}}{6}.$$

ONE OR MORE CORRECT ANSWERS

20. Range of the function $f(x) = \sqrt{\sin x} - \sqrt{\cos x}$
 is $[-1, 1]$
 and the function is periodic with period 2π

21. If $\frac{\pi}{2} < x < \pi$, no root. It can be seen that
 $x = \frac{\pi}{4}$ is the only real root.

22. Given equation is $\sin(x + x^2) - \sin x^2 = \sin x$
 solving we get $x^2 + x = 2\pi, n \in \mathbb{W}$ or
 $x^2 = 2p\pi, p \in \mathbb{W}$

23. Draw the graphs of the functions
 $f(x) = -2\sqrt{3}\pi \sin x$ and
 $g(x) = |x + \pi| + |x - 2\pi|$. The two graphs
 intersect at exactly four distinct points.

24. $2 \cos \theta (1 - \sin \varphi) = \frac{2 \sin^2 \theta}{\sin \theta} \cos \varphi - 1$

$$= 2 \sin \theta \cos \varphi - 1$$

$$2 \cos \theta - 2 \cos \theta \sin \varphi = 2 \sin \theta \cos \varphi - 1$$

$$2 \cos \theta + 1 = 2 \sin(\theta + \varphi)$$

$$\tan(2\pi - \theta) > 0 \Rightarrow \tan \theta < 0 \text{ and}$$

$$-1 < \sin \theta < \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta \in \left(\frac{3\pi}{2}, \frac{5\pi}{3}\right)$$

$$\frac{1}{2} < \sin(\theta + \varphi) < 1,$$

TRIGONOMETRIC EQUATIONS

JEE ADVANCED - VOL - II

$$\left(\frac{3\pi}{2} < \theta < \frac{5\pi}{3}, \frac{1}{2} < \frac{2\cos\theta + 1}{2} < 1 \right)$$

$$\Rightarrow 2\pi + \frac{\pi}{6} < \theta + \varphi < \frac{5\pi}{6} + 2\pi$$

$$2\pi + \frac{\pi}{6} - \theta_{\max} < \varphi < 2\pi + \frac{5\pi}{6} - \theta_{\min}$$

$$\frac{\pi}{2} < \varphi < \frac{4\pi}{3}$$

25. The equation is possible only when each term is unity, i.e;
 $\sin 2x = 1, \sin 3x = 1, \sin 4x = 1, \dots, \sin nx = 1$
 if $\sin 2x = 1$

$$2x = \frac{(4n+1)\pi}{2}$$

$$\Rightarrow x = \frac{(4n+1)\pi}{4}$$

$$\text{then } \Rightarrow 3x = \frac{(4n+1)3\pi}{4}$$

$$\Rightarrow \sin 3x \neq 1$$

$$\Rightarrow (b), (d) \text{ are correct}$$

26. Conceptual

27. The given equation can be written as

$$\begin{aligned} \sqrt{(\cos^2 x - \sin^2 x)} + \sqrt{(\cos x + \sin x)^2} \\ = 2\sqrt{(\cos x + \sin x)} \end{aligned}$$

$$\Rightarrow \sqrt{(\cos x + \sin x)} \sqrt{(\cos x - \sin x)} + \sqrt{(\cos x + \sin x)} = 2\sqrt{(\cos x + \sin x)}$$

$$\Rightarrow \text{Either } \cos x + \sin x = 0$$

$$\Rightarrow \tan x = -1$$

$$\Rightarrow x = n\pi - \frac{\pi}{4} \quad (n \in I)$$

$$\text{or } \sqrt{(\cos x - \sin x)} + \sqrt{(\cos x + \sin x)} = 2$$

$$\Rightarrow 2\cos x + 2\sqrt{(\cos^2 x - \sin^2 x)} = 4$$

$$\Rightarrow \cos^2 x - \sin^2 x = (2 - \cos x)^2$$

$$\Rightarrow \cos^2 x + 4\cos x - 5 = 0$$

$$\Rightarrow \cos x = \frac{-4 \pm \sqrt{(16+20)}}{2} = -5 \text{ or } 1$$

$$\text{But } \cos x \neq -5 \text{ or } \cos x = 1$$

$$\Rightarrow x = 2n\pi, n \in I.$$

28. On putting $\sin\left(x - \frac{\pi}{3}\right) = t$, we get

$$2t^2 - 5t + 2 > 0$$

$$(t-2)(2t-1) > 0$$

$$\text{Since } t-2 < 0$$

$$\text{The last inequation is equivalent to } 2t-1 < 0$$

$$\text{or } \sin\left(x - \frac{\pi}{3}\right) < \frac{1}{2}$$

$$\Rightarrow (d) \text{ is correct.}$$

$$\text{Now } \sin\left(x - \frac{\pi}{3}\right) < \frac{1}{2}$$

$$\Rightarrow 2n\pi - \frac{7\pi}{6} < x - \frac{\pi}{3} < 2n\pi - \frac{\pi}{6}$$

(See figure)

$$\Rightarrow \left(\frac{12n-5}{6}\right) < x < \frac{(4n+1)\pi}{2}$$

$$\Rightarrow (c) \text{ is correct.}$$

The solution given in choice (b) follows from general solution $\tan = 0$

Hence (b), (c), (d) are correct.

29. The given equation is possible only when

$$\cos ax = 0 \quad (\because \sin x \leq 1)$$

$$\text{When } \cos ax = 0, \quad \sin x = 1$$

$$\Rightarrow ax = \frac{(2n+1)\pi}{2}, x = \frac{(4m+1)\pi}{2}$$

$$\text{From first equation, } x = \frac{(2n+1)\pi}{2a}$$

$$\text{Now } \frac{(2n+1)\pi}{2a} = \frac{(4m+1)\pi}{4m+1}$$

$$\Rightarrow a = \frac{2n+1}{4m+1} \quad \dots(i)$$

$\Rightarrow$ a has to be rational since $\frac{2n+1}{4m+1}$ is rational.

$\Rightarrow$ (a) is false, (b) is true.

Now (c) will be correct if

$$\frac{(2n+1)\pi}{2a} = \frac{(2009)\pi}{2}$$

$$\Rightarrow a = \frac{2n+1}{2009}$$

Take $n = 1004$

then $a = 1$

and from (i) $4m+1 = 2n+1$

$$\Rightarrow 4m = 2008$$

$$\Rightarrow m = 502$$

$\Rightarrow$ Thus

$$\Rightarrow \frac{2009\pi}{2}$$

is a solution for $a = 1$.

$\Rightarrow$ (c) is true and (d) is false.

$$30. \quad \sin x = \frac{1 + \sqrt{1+4a}}{2}$$

We must have $1 + 4a \geq 0$

$$\text{and } 1 \leq \frac{1 + \sqrt{1+4a}}{2} \leq +1$$

The above two conditions are equivalent to

$$-\frac{1}{4} \leq a < 0.$$

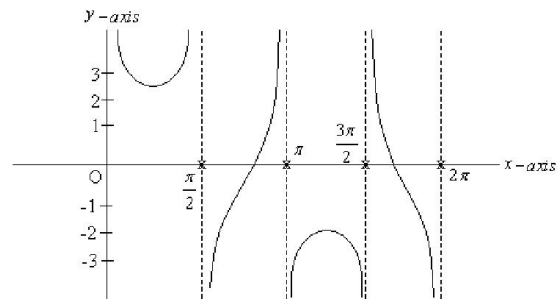
For every a in this interval, the equation will give four values of x [Since over $(0, 2\pi)$ any value by $\sin x$ is attained for two values of x and the quadratic is giving two values of $\sin x$]

COMPREHENSION PASSAGES

31, 32: $|x| \leq 2\pi$ and $\sqrt{4\pi^2 - x^2} \in [0, 2\pi]$. Put $\cos \sqrt{4\pi^2 - x^2} = t$, $|t| \leq 1$ if $a = 1$ then $\tan t$

$$= 2. \text{ If } a = -\frac{1}{2} \text{ then } \tan t = -1$$

Passage - 2 33, 34, 35



$$1) \quad a > 2\sqrt{2} \quad (\text{or}) \quad a < -2\sqrt{2}$$

36. Graphs of $y = \sin x$ and $y = \frac{|x|}{10}$ meet exactly six times. Hence, No. of solutions = 6.

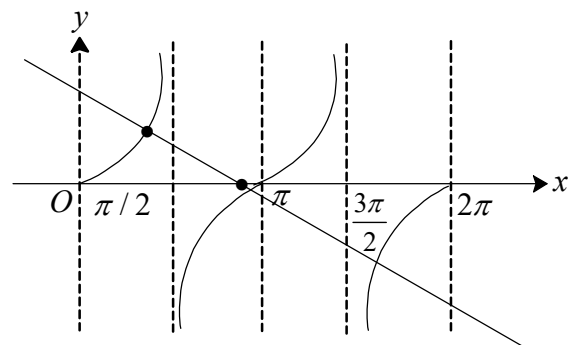
$$37. \quad 3x + 2 \tan x = \frac{5\pi}{2}$$

$$\Rightarrow \tan x = \frac{5\pi}{4} - \frac{3x}{2}$$

Graphs of $y = \frac{5\pi}{4} - \frac{3x}{2}$ and $y = \tan x$, meet

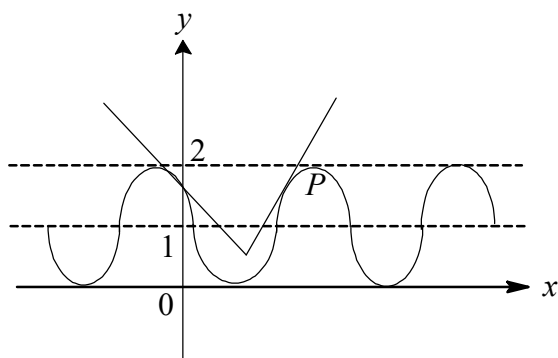
exactly three times in $[0, 2\pi]$.

Thus no. of solutions = 3.



38. Slope of $y = 1 - \sin x$ is $-\cos x$. Slope of

$$y = \frac{\sqrt{3}}{2} \left| x - \frac{\pi}{2} \right| + a \text{ for } x > \frac{\pi}{2} \text{ is } \frac{\sqrt{3}}{2}.$$



$$\Rightarrow \cos x = -\frac{\sqrt{3}}{2} \Rightarrow x = \frac{7\pi}{6} \Rightarrow p = \left(\frac{7\pi}{6}, \frac{3}{2}\right)$$

If $y = \frac{\sqrt{3}}{2} \left|x - \frac{\pi}{2}\right| + a$, passes through 'p'

$$\text{then } a = \frac{3}{2} - \frac{\pi}{\sqrt{3}}$$

$$\Rightarrow a > \frac{3}{2} - \frac{\pi}{\sqrt{3}}$$

MATRIX MATCHING

39. a) $2 - p \geq 0$ and $p \geq 0$ and $\sqrt{2} + \sqrt{2 - p} \leq \sqrt{4 + p}$

b) Solving we get $\sin x = \frac{a-4}{2}, \left|\frac{a-4}{2}\right| \leq 1$

c) The given equation is $4t^2 + 2t - 1 = a, t = \cos \frac{x}{2}$

d) The given equation is $t^2 - 2t - 2a, t = \sin 2x$

40. (A) $\because \sin x = -\frac{1}{2}$

$$= -\sin \frac{\pi}{6}$$

$$= \sin \left(\pi + \frac{\pi}{6}\right), \sin \left(2\pi - \frac{\pi}{6}\right)$$

$$\therefore x = \frac{7\pi}{6}, \frac{11\pi}{6} \quad \dots\dots(i)$$

$$\text{and } \cos x = -\frac{\sqrt{3}}{2} = -\cos \frac{\pi}{6}$$

$$= \cos \left(\pi - \frac{\pi}{6}\right), \cos \left(\pi + \frac{\pi}{6}\right)$$

$$\therefore x = \frac{5\pi}{6}, \frac{7\pi}{6} \quad \dots\dots(ii)$$

From Eqs. (i) and (ii). It is clear tht

$$\alpha = \frac{7\pi}{6}, \beta = \frac{11\pi}{6}, \gamma = \frac{5\pi}{6}$$

$$\Rightarrow \alpha + \beta = 3\pi(S), \beta - \gamma = \pi(Q)$$

(B) $\therefore \cot x = -\sqrt{3}$

$$= -\cot \frac{\pi}{6}$$

$$= \cot \left(\pi - \frac{\pi}{6}\right), \cot \left(2\pi - \frac{\pi}{6}\right)$$

$$\therefore x = \frac{5\pi}{6}, \frac{11\pi}{6} \quad \dots\dots(i)$$

$$\text{and } \operatorname{cosec} x = -2 = -\operatorname{cosec} \frac{\pi}{6}$$

$$= \operatorname{cosec} \left(\pi + \frac{\pi}{6}\right), \operatorname{cosec} \left(2\pi - \frac{\pi}{6}\right)$$

$$\therefore x = \frac{7\pi}{6}, \frac{11\pi}{6} \quad (C) \because \sin x = -\frac{1}{2}$$

$$= -\sin \frac{\pi}{6}$$

$$= \sin \left(\pi + \frac{\pi}{6}\right), \sin \left(2\pi - \frac{\pi}{6}\right)$$

$$\therefore x = \frac{7\pi}{6}, \frac{11\pi}{6} \quad \dots\dots(i)$$

and $\tan x = \frac{1}{\sqrt{3}}$

$$= \tan \frac{\pi}{6}$$

$$= \tan \frac{\pi}{6}, \tan \left(\pi + \frac{\pi}{6} \right)$$

$$\therefore x = \frac{\pi}{6}, \frac{7\pi}{6} \quad \dots\dots(ii)$$

From Eqs. (i) and (ii), it is clear that

$$\alpha = \frac{7\pi}{6}, \beta = \frac{11\pi}{6}, \gamma = \frac{\pi}{6}$$

$$\Rightarrow \alpha + \beta = 3\pi, (S), \beta + \gamma = 2\pi(T), \alpha - \gamma = \pi(R) \quad \dots\dots(ii)$$

From Eqs. (i) and (ii), it is clear that

$$\alpha = \frac{11\pi}{6}, \beta = \frac{5\pi}{6}, \gamma = \frac{7\pi}{6}$$

$$\Rightarrow \beta + \gamma = 2\pi(T), \alpha - \beta = \pi(P).$$

INTEGER ANSWER TYPE

41. The number of distinct real root of

$$\tan \left(\frac{\pi}{2} \cos x \right) = \cot(\pi \sin x), |x| \leq \frac{\pi}{2} \text{ is}$$

42. Clearly $x \neq 0$. $\frac{2\pi x}{x^2 + x + 1} = n\pi - \frac{\pi}{3}, n \in \mathbb{Z}$

$$\Rightarrow x + \frac{1}{x} = \frac{7-3n}{3n-1}$$

$$\Rightarrow n = 0, \pm 1$$

43. given that $(x-6)^2 + 3 = 3 \sin \frac{\pi a}{x}$

$$\therefore a = 12n + 3, n \in \mathbb{Z}$$

44. $y \geq 0$ and $\pi - \sqrt{y} > 0$ and $\pi - \sqrt{y} \neq 1$

$$\therefore 0 \leq \sqrt{y} < \pi. \text{ Also } x > 0 \text{ and } x \neq 1$$

$$\pi - \sqrt{y} = x \text{ and solve}$$

45. $\sin 3x = -1$ and $\cos 2x = -1$

$$\therefore x = n\pi + (-1)^n \frac{\pi}{2}, n \in \mathbb{Z}$$

$$46. \sin(2 \sin x) = \sin \left(\frac{\pi}{2} - 2 \cos x \right)$$

$$\sin x + \cos x = \frac{\pi}{4}$$

s.o.b.s

$$1 + \sin 2x = \frac{\pi^2}{16}$$

$$\sin 2x = \frac{\pi^2 - 16}{16}$$

$$\therefore \tan x + \cot x = \frac{2}{\sin 2x} = \frac{2 \times 16}{\pi^2 - 16} = \frac{32}{\pi^2 - 16}$$

$$\therefore a = 32, b = 16, c = 2$$

$$\frac{a+b+c}{25} = 2$$

$$47. [x] = \frac{5 \pm \sqrt{25 + 4 \sin x - 24}}{2.1}$$

$$= \frac{5 \pm \sqrt{1 + 4 \sin x}}{2}$$

$$-1 \leq \sin x \leq 1$$

$$-4 \leq 4 \sin x \leq 4$$

$$-3 \leq 1 + 4 \sin x \leq 5$$

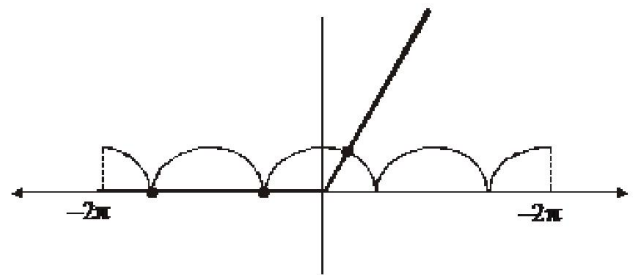
$$0 \leq 1 + 4 \sin x \leq 5$$

$$\Rightarrow [x] \text{ is an integer} \Leftrightarrow \sin x = 0$$

$$\Rightarrow [x] = 3$$

$$\Rightarrow x = \pi$$

48. Check from graph



49. $(y+z) \cos 3\theta - (xyz) \sin 3\theta = 0$
.....(1)

TRIGONOMETRIC EQUATIONS

$$xyz \sin 3\theta = 2(\cos 3\theta)z + (2 \sin 3\theta)y$$

.....(2)

$$\therefore (y+z) \cos 3\theta = (2 \cos 3\theta)z$$

$$+(2 \sin 3\theta)y = (y+2z) \cos 3\theta + y \sin 3\theta$$

$$y(\cos 3\theta - 2 \sin 3\theta) = z \cos 3\theta \text{ and}$$

$$y(\sin 3\theta - \cos 3\theta) = 0$$

$$\Rightarrow \sin 3\theta - \cos 3\theta = 0 \Rightarrow \sin 3\theta = \cos 3\theta$$

$$\therefore 3\theta = n\pi + \pi/4$$

50. $\tan \theta = \cot 5\theta$

$$\Rightarrow \cos 6\theta = \theta$$

$$4 \cos^3 2\theta - 3 \cos 2\theta = 0$$

$$\Rightarrow \cos 2\theta = 0 \text{ or } \pm \frac{\sqrt{3}}{2}, \sin 2\theta = \cos 4\theta$$

$$\Rightarrow 2 \sin^2 2\theta + \sin 2\theta - 1 = 0$$

$$2 \sin^2 2\theta + 2 \sin 2\theta - \sin 2\theta - 1 = 0$$

$$\sin 2\theta = 0 \text{ and } \sin \theta = \frac{1}{2}$$

$$\cos 2\theta = 0 \text{ and } \sin 2\theta = -1$$

$$\Rightarrow 2\theta = -\frac{\pi}{2} \Rightarrow \theta = -\frac{\pi}{4}$$

$$\cos 2\theta = \pm \frac{\sqrt{3}}{2}, \sin 2\theta = \frac{1}{2}$$

$$\Rightarrow 2\theta = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow \theta = \frac{\pi}{12}, \frac{5\pi}{12}$$

$$\therefore = -\frac{\pi}{4}, \frac{\pi}{12}, \frac{5\pi}{12}$$